

CLIENT حكومة دبي GOVERNMENT OF DUBAI	CONSULTANT PARSONS	CONTRACTOR 中国土木 CCECC										
METHOD STATEMENT SUBMITTAL												
PROJECT: R1181/2 Improvement of Al Safa Road												
FOR INFORMATION AND RECORDS	FOR APPROVAL	 Reference No. : CCECC-R1181-2-CC-UT-MSRA-022 Rev. No. : 00 Date : 24-Mar-26										
<table style="width:100%;"> <tr> <td style="width:50%; vertical-align: top;"> To: Parsons Overseas Limited P.O.Box 9123 Dubai - U.A.E. </td> <td style="width:50%; vertical-align: top;"> From: China Civil Engineering Construction Corporation P.O.Box 393249 Dubai, U.A.E. Originator: Mohsen D Gazzaz Project Director </td> </tr> </table>			To: Parsons Overseas Limited P.O.Box 9123 Dubai - U.A.E.	From: China Civil Engineering Construction Corporation P.O.Box 393249 Dubai, U.A.E. Originator: Mohsen D Gazzaz Project Director								
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Document Title:- METHOD STATEMENT AND RISK ASSESSMENT FOR DM - STORM WATER NETWORK Location of related works:- As per Approved Shop Drawings Date of Approval Required by: ASAP		 										
Other Remarks: <table style="width:100%;"> <tr><td style="width:5%;">1</td><td>Method Statement (MS) Attached.</td></tr> <tr><td>2</td><td>Risk Assessment (RA) Attached.</td></tr> <tr><td>3</td><td>Inspection & Test Plan (ITP) Attached.</td></tr> <tr><td>4</td><td>Checklist Attached.</td></tr> <tr><td>5</td><td>Emergency Contact Details Attached.</td></tr> </table>			1	Method Statement (MS) Attached.	2	Risk Assessment (RA) Attached.	3	Inspection & Test Plan (ITP) Attached.	4	Checklist Attached.	5	Emergency Contact Details Attached.
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CLIENT/ENGINEERS RESPONSE (Please tick relevant box)												
Clients / Engineer's Comments: Risk Assessment part approved as comments - Comments sheet attached. → - subject to full compliance with Project specification and DM standards. - work to be done in coordination with DM and Parsons site team. 1/4/2026		 ☐ Approved ☒ Approved as Noted ☐ Revise & Resubmit ☐ No Approval Required										
For Client / Engineer Representative:												
Name: Project Director Signature:	Reference: Date:											

DISTRIBUTION		FILE NO:		
CCECC				
PROJECT: RTA (R1181/2 - IMPROVEMENT OF AL SAFA STREET				
DOCUMENT / CORRESPONDENCE DISTRIBUTION				
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HSE		✓		
TRAFFIC				✓
DESIGN / TECHNICAL				✓
QA/QC / MATERIAL	✓			
NOC				✓
CONSTRUCTION BRIDGE				✓
CONSTRUCTION ROAD				✓
CONSTRUCTION UTILITY		✓		
SURVEYOR				
STORE				
HEAD OFFICE				
SUBCON				
OTHERS				


30/4/26

SH&E Review Comments

Project Name:	Improvement of Al Safa Road	Project #	R1181/2	Date	30.03.2026
Contractor:	China Civil Engineering Construction Corporation	Reviewer	Ghulam Moinuddin HSE Engineer (Parsons)	Sign:	
Document Title:	Method Statement & Risk Assessment for DM-Storm Water Network	Document #	CCECC-R1181/2-CC-ST-MSRA-022	Rev	00
Status	B = Approved with Comments Status A = Approved, Status B = Approved with Comments, Status C = Revise and Resubmit, Status D = Rejected				

Title / Topic	Ref	Parsons's Review / Comments	Contractor's Inputs	Ref	Parsons Status
		The submitted MSRA is Approved with comments, subject to incorporation of the attached review comments. The Contractor shall ensure all comments are addressed and implemented prior to commencement of the activity			
General		Compliance of all applicable local authorities' approvals and NOCs, Drawings layout, TRA QHSS MSTG, WZTMM, RTA Circulars and DM Code of Construction Safety Practice and contractual requirements. Relevant authority requirements such as RTA WZTM, ITS authority NOC, Dubai Municipality Code of Construction Safety Practice and utility authority procedures shall be clearly referenced and implemented.			
Working Near Live Traffic	Entire Document	Specific Risk Assessment and control measures for working near live traffic shall be included as per RTA WZTM requirements. Provision of concrete barriers, trained flagmen, signage, and lighting shall be ensured.			C
Utilities Protection	Sec 4.1 / 4.2	Method Statement shall clearly include utilities detection, scanning records, trial pits, and NOC compliance prior to excavation.			C
Confined Space (Manholes)	Sec 4.7	Confined space hazards are not addressed. Include permit system, gas testing, ventilation, standby rescue team, and emergency arrangements.			C
Environmental Requirements	Entire Document	Environmental controls including dust suppression, noise control, waste management, and water discharge shall be detailed in line with CEMP and DM requirements.			C
Permit to Work System	Entire Document	All activities shall be carried out under approved Permit to Work system (Excavation, Confined Space, Lifting, etc.).			C
Plant & Equipment	Sec 3.2	All plants and equipment shall have valid third-party inspection certificates, operator competency certificates, and QR code system implemented as per project requirements.			C

(72)

UTILITIES WORKS

METHOD STATEMENT AND RISK ASSESSMENT FOR DM - STORM WATER NETWORK



HISTORY

Referential		Submission :		☐ For Information Only ☒ For Approval	
Rev	Date	Modifications Content		Writer	Function
00	24-Mar-26	First Issue		Jeevanantham.R	QA/QC Engineer

CCECC REVIEW AND VERIFIED BY :

	REVIEWER 1	REVIEWER 2	REVIEWER 3	REVIEWER 4	REVIEWER 5	VERIFIED BY
Name	Ahmed Ghanem	Syed Abdul Moid	Mirza Shahabuddin Baig	Ahmed Abdalla	Ahmed Elkadi	Mohsen D Gazzaz
Entity / Function	Construction Manager	QA/QC Manager	HSE Manager	Technical Manager	Deputy. Project Manager	Project Director
Date	24-Mar-26	24-Mar-26	24-Mar-26	24-Mar-26	24-Mar-26	24-Mar-26
Signature						

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METHOD STATEMENT FOR STORM WATER WORKS

1. Purpose and Scope

1.1 Purpose

The purpose of this method statement is to ensure the Implementation of the systematic an effective Storm water network in this project and all its related activities to be executed as per DM standards and procedures for this project and to ensure that all works are carried out in accordance with the Approved shop drawings, Health, Safety, Environment requirements and Contract Documents. This Method statement outline the procedures to be followed in implementing the Storm water network are carried out in accordance with the Approved Shop Drawings and Project Specifications.

1.2 Scope

The scope of this work includes the Laying of the GRP pipelines with GRP fittings, uPVC Pipelines (solid and perforated) with all necessary chambers along with concrete surrounding works, transportation and installation of chambers for the Project. All the works shall be supervised by the Engineers Representative and the Site Engineer for the

Project: **R1181-2 - IMPROVEMENT OF AL SAFA ROAD.**

2. Definitions and References

2.1 Definitions

Client	Roads & Transport Authority (RTA)
Consultant	Parsons Overseas Limited (POL)
Contractor	China Civil Engineering Construction Corporation – Dubai (CCECC)
DEWA	Dubai Electricity & Water Authority
DM	Dubai Municipality

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2.2 Document References

ISO 9001:2015	Quality Management Systems - Requirements
ISO 45001:2018	Occupational Health and Safety Management System - Requirements
ISO 14001:2015	Environmental Management System - Requirements
Technical Specifications	Project Specifications
PAS 128:2014	Specification for underground utility detection, verification, and location.
Dubai Municipality Control Stations	As and where applicable
Dubai Code of Construction Safety Practice	As and where applicable
RTA / DM Requirements	RTA Work Zone Traffic Management Manual , DMCO CSP, MSTG & RTA Circulars.
Federal Law No. (24) Of 1999	Concerning the Protection, Development and Amendments of the Environment (amended in Law No. (11) Of 2006)
DM NRCS Technical Guideline	Translocation, Handling, and Restoration of Wildlife (DM-ED-GU-01-THRWTG).
DEWA	Water Technical Standards and Specifications & DEWA Handbook on EHV Over Headline & Underground cable protective
IS 1377 Part 9: 1990	Methods of Test for Civil Engineering Purpose
BS 5930: 1999	Codes of practices for site investigation
CCECC-R1181-2-CC-DS-001	Project Quality Plan (PQP)
CCECC-R1181-2-CC-DS-002	Health and Safety Construction Management Plan (
CCECC-R1181-2-CC-DS-003	Construction Environment Management Plan (CEMP)
CCECC-R1181-2-CC-DS-005	Construction Waste Management Plan (CWMP)
CCECC-R1181-2-CC-DS-007	Construction Lifting Management Plan (CLMP)
CCECC-R1181-2-CC-DS-006	Traffic Management Plan (TMP) (Where as applicable)
CCECC-R1181-2-CC-DS-010	Emergency / Contingency & Business Continuity Plan (ECBCP)

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3. Process Resources

3.1 Personnel Roles and Responsibilities

3.1.1 Project Director

- Overall, in-charge of the project and company representative.
- Responsible for project delivery within time, cost and quality requirements.
- Liaise with Client, Consultant and Authorities.
- Ensure implementation of Project Quality Plan (PQP) and HSE Plan.
- Monitor overall progress and resolve critical issues.

3.1.2 Deputy Project Manager

- Manage day-to-day site operations.
- Ensure works are executed as per approved drawings, specifications and method statements.
- Monitor progress and coordinate inspections as per ITP.
- Ensure availability of approved materials and resources.
- Supervise construction team and subcontractors.

3.1.3 Construction Manager

- Plan and supervise construction activities as per approved programme.
- Coordinate manpower, plant and materials.
- Ensure compliance with quality, safety and contractual requirements.
- Monitor subcontractor performance.
- Stop unsafe work if required.

3.1.4 Technical Manager

- Manage shop drawings, technical submittals and as-built documentation.
- Coordinate with service authorities for approvals.
- Raise and resolve RFIs and technical discrepancies.
- Maintain submittal logs and document control.

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3.1.5 QAQC Manager

- Implement and monitor the Project Quality Plan.
- Review and approve ITPs, Method Statements and Material Submittals.
- Ensure inspections are conducted as per approved procedures.
- Issue and close NCRs and coordinate with Consultant on quality matters.

3.1.6 QAQC Engineer

- Verify works comply with approved drawings and specifications.
- Review RFAs and coordinate inspections.
- Monitor testing activities and maintain quality records.
- Ensure only approved materials are used on site.
- Monitoring each site activity followed as per approved drawings and approved MSRA.

3.1.7 Material Engineer / Lab Technician

- Ensure all permanent materials are approved prior to use.
- Monitor material testing and maintain testing logs.
- Verify calibration of testing equipment.
- Coordinate with approved laboratories for required tests.
- Evaluate test results before submission to consultant.
- Co-ordinating with DCL and Engineer / Consultant for all Material testing of the Project.
- Maintaining the Concrete Checkpoint and site lab as per the requirement.

3.1.8 HSE Manager / HSE Officer

- Overall responsible for implementation of Project HSE Plan.
- Conduct risk assessments and regular safety inspections.
- Ensure compliance with statutory and environmental regulations.
- Investigate incidents and implement corrective actions.
- Stop unsafe activities when required.
- Verify work permits and high-risk activity controls.
- Conduct toolbox talks and safety briefings.
- Report unsafe conditions and enforce corrective actions.

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3.1.9 Traffic Management Team (TTM Manager / Engineer)

- Prepare and implement approved Traffic Diversion Plans.
- Coordinate with Authorities for traffic permits and approvals.
- Ensure installation and maintenance of traffic control devices.
- Monitor traffic safety compliance at site.
- Maintain traffic management records and reports.

3.1.10 Site Engineer/ Foreman / Chargehand

- Supervise daily construction activities.
- Ensure work is carried out as per approved method statements and permits.
- Coordinate with consultant for inspections and approvals.
- Monitor manpower, materials and equipment usage.
- Ensure HSE compliance at site.

3.1.11 Chief Surveyor / Surveyor

- Carry out setting-out works as per approved drawings.
- Maintain survey control points and calibrated instruments.
- Prepare and submit as-built survey records.
- Ensure dimensional accuracy of executed works.

3.1.12 Workers, Operators & Drivers

- Perform assigned tasks as per instructions and approved procedures.
- Use appropriate PPE at all times.
- Operate only authorized equipment with valid licenses.
- Follow basic guidelines and HSE Requirements as provided in this document.
- Conduct daily equipment checks.
- Report unsafe conditions immediately.

3.1.13 Subcontractor (If Applicable)

- Follow Project Specification and Contract Drawings requirements.
- Follow all the Quality and Safety requirements in the Project.
- Have a safe site, plant, structures and ways of working.
- Make sure plant, structures and substances are used, handled and stored safely.
- Provide facilities for the welfare of workers, such as drinking water and toilets.

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- Provide the necessary information, training and supervision to protect people from risk.
- Monitor health and workplace conditions.

3.2 Tools and Equipment

- Shovel
- Excavator / JCB
- Wheel Loader / Boom Loader
- Trucks
- Water Tanker
- Baby Roller
- Plate Compactor
- Generators
- Crane Mounted Trucks (Hi-ab).
- Mobile Crane
- Gauge with tubes
- Concrete Vibrators
- Loading/Un loading belts
- Survey Equipment's
- Measuring Tape
- FDT & Concrete testing equipment.

Note: -

- 1 The above listed equipment or additional equipment shall be mobilized subject to the requirement as we need necessary.
- 2 Calibration Certificates will be available as applicable for each instrument used on site.
- 3 All Machineries, Equipment's shall be "Third Party Inspected and Certified.
- 4 All Machineries, Equipment's shall be labeled with QR Code and Maintain the Color Code.

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3.3 Material

- GRP Pipes & Fittings
- uPVC Pipes & Fittings
- Geotextile
- Granular Bedding Material
- Water proofing materials
- Polythene Sheet
- Approved Ready Mix Concrete
- Approved Reinforcement Steel
- Approved Backfilling Material
- Warning Tapes
- Chemicals (COSHH assessment will be conducted prior to work)
- Plastic barriers, Concrete barriers, Cones, Ropes, Warning tapes & Warning signs.

3.4 Manpower

- Construction Manager
- Site Engineer
- Safety Officer
- Site Supervisor
- Foremen's
- Chargehand's
- Carpenters
- Masons
- Electrician
- Operator's
- Driver's
- Flagman's

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3.5 Transportation

- For all Vehicles / equipment's to be used on site safety equipment such as fire extinguisher, first aid kit, vehicle emergency equipment and communication devices will be providing. The operators shall be in possession of necessary licenses and third part certificates of competence of the equipment's to be used.
- Site speed limit should be 20 km/h and should not exceed in any case.

4. Procedure

4.1 Prior Activities

- Before commencing any work, where necessary, a notification will be sent to the concerned authorities by Stakeholder Manager whose assets can be found on site according to the information available where required.
- Construction NOC to be obtained from the Authorities before starting the work at site.
- Preparations activities will start only after formal instruction from the Construction Manager to the site team, after the approvals at NOC. No work shall commence without NOC.
- Internal traffic arrangements will be made in the work site area with the assistance of safety personnel.
- A representative of the Engineer's organization will witness, observe and record the works as per approved drawings, and a Method Statement and ensure that all the operations comply with the contract specifications.
- A responsible Engineer & Foreman in charge of excavation works will ensure that all utilities in the area are identified.
- Existing services will be located with the available as-built drawings and scanning the relevant area by cable detectors.
- Obtain approval for all related documentation including the Permit to Work, the relevant Method Statement, Shop Drawings, and the Pre-Qualification for the Subcontractor.

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- Conduct the necessary safety induction for employees.
- Inspection Request shall be submitted upon completion of each activity for Engineers inspection and approval.

4.2 Protection of Existing Services:

- Existing Services if located will be protected as per the approved norms of the Engineer / authority. Once the work is completed, the services will be reinstated back. Backfilling will proceed only after the reinstatement is approved by the Engineer / authority.

4.3 Execution Procedure:

4.3.1 Site Planning & Preparation

- All site team involved in the execution of the work will be briefed on the contents of this method statement.
- Benchmarks will be established, using approved control points by the surveyor and approved by the consultant.
- The Surveyor will establish the correct location and setting out to start the work as per approved shop drawing.
- A permit to work (PTW) will be obtained from the HSE by the site engineer in accordance with the approved Permit to Work System as per HSE Plan.
- The setting out drawing for the Stormwater lines are prepared taking into consideration the duly approved service corridor reservation, the diameter of the pipe, the invert levels, the locations of the chambers, the vertical profiles with all interpolations of interfering services.
- The prepared setting out drawing and the vertical profile drawings shall be approved by the consultants and proper alignment assessments shall be done to ascertain the smooth supply network and to ease maintenance facility.

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4.3.2 Arrangement of Access

- Safe access to site shall be made possible by applying and acquiring entry passes to the staff, equipment and transport. Safe road access provided on site shall be used as a safety precaution to prevent unnecessary damage to existing services or facilities on site.
- The traffic diversion, if required, shall be carried out as per department concerned approval.
- All the necessary signboards, signals and flashlights shall be provided as per requirement.
- All personnel protective equipment must be available and worn by the personnel working at site.

4.3.3 Setting Out

- The works will be set out in accordance with the agreed procedure & approval by consultant. The setting out will be carried out from Temporary Benchmarks / survey stations established at site in concurrence with the Consultant. The setting out shall be carried out in accordance with the approved construction drawings.

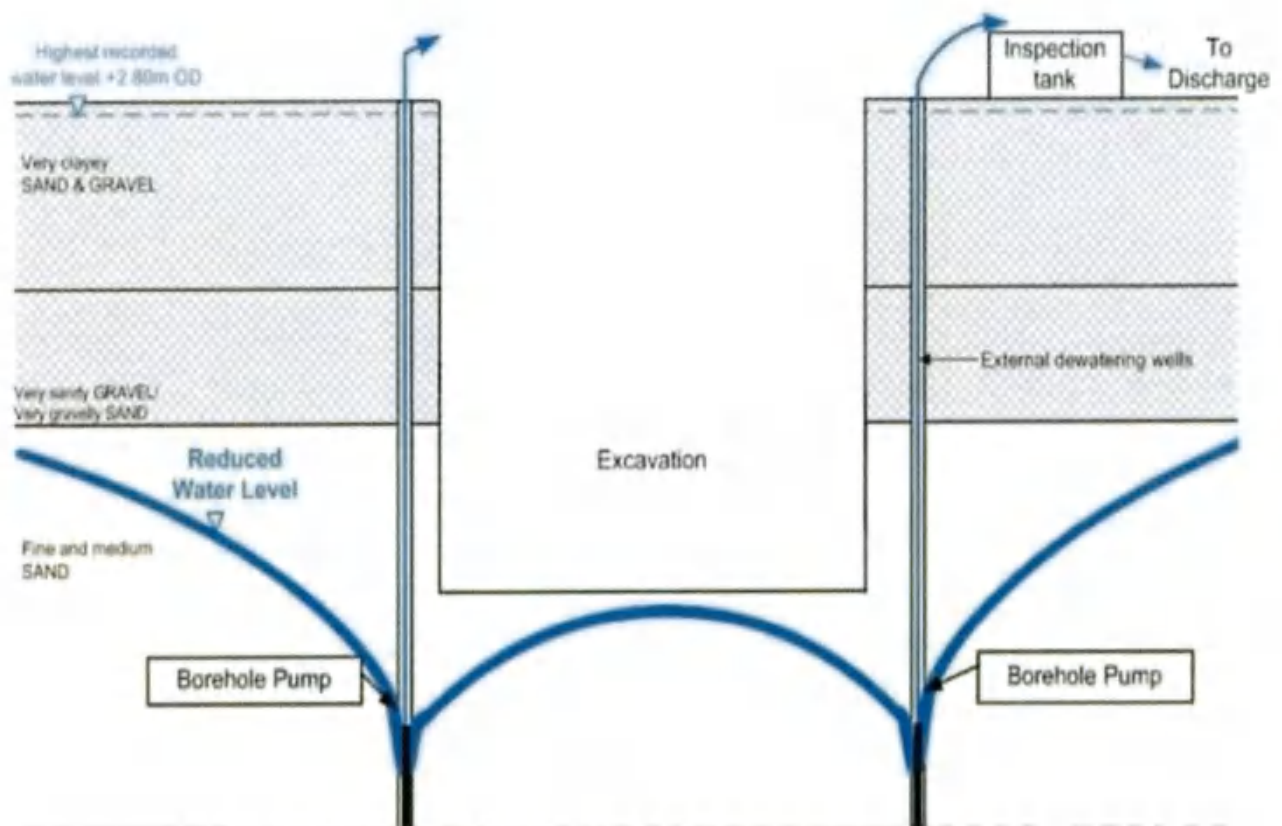
4.3.4 Dewatering (If required)

- Where ground water is encountered, dewatering will be done by means of an efficient deep well system. This includes adequate number of wells drilled either on one side or both sides of the trench, depending upon the amount of ground water to be pumped. Wells will be drilled on approved locations (if required) to determine the water table level for dewatering purposes.
- The water level will be drawn down to a minimum of 1m below the bottom of the excavation to maintain the undisturbed state of the foundation soil. The ground water pumped will be discharged at a suitable and approved location. Dewatering method statement will be submitted separately prior to starting work at site.
- Obtain all necessary permits & NOCs from respective authorities and service providers required for site activities.

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- In any failure occur in the dewatering system, the water shall be removed by approved tanker services till the flooded water remove and to start standby dewatering system.
- Works shall not be permitted to continue in such cases.
- All necessary shop drawings shall be submitted, and work shall start only after approval is obtained.

Cone OF Depression



Deep Well Dewatering System

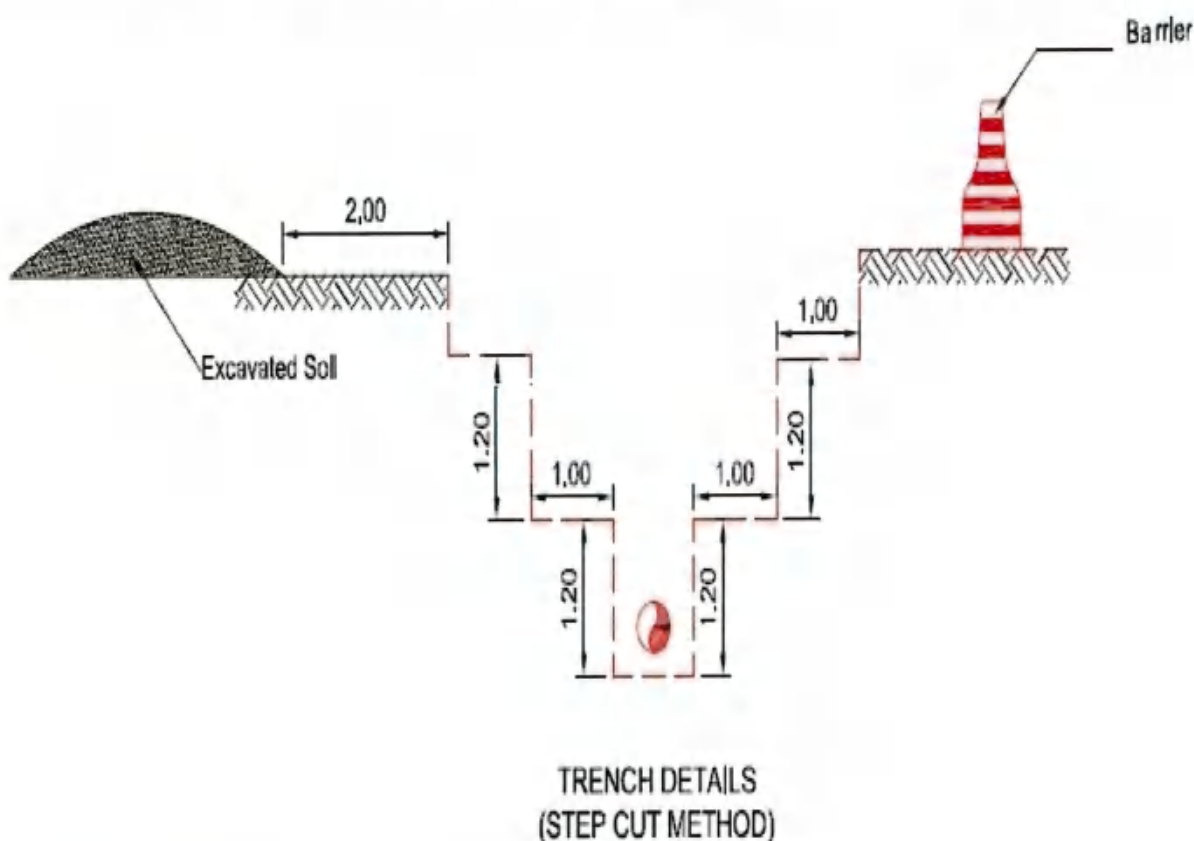
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4.3.5 Excavation & Formation Preparation:

- Ensure that the excavations are started after obtaining required New Excavation permits.
- Existing services through the Storm water line route shall be exposed manually.
- Trenches for storm water lines shall be excavated to sufficient depth and width, and as per
- approved shop drawing / cross section to enable the pipe and any specified or agreed joint bedding and surrounding to be accommodated.
- Minimum trench width shall be equal to $1.5 \times \text{Dia} + 500\text{mm}$ subject to minimum trench width of 750mm as per the Dubai Municipality standard cross section.
- Excavation for the trench shall be carried out with excavator as per the typical cross section provided that adequate spacing from the existing utilities is available at site.
- Final slopes of the trench walls and number of steps shall be determined as per existing soil conditions and depth of the trench
- Ensure that the excavated material shall be stocked minimum 2m away from the edges of the trench.
- Excavation shall be carried out under the supervision of the foreman and site engineer.
- Ensure the proper safe access to the trench by staircase shall be provided wherever possible, if not the proper suitable access shall be provided, the same shall be agreed with the Engineer
- If the ladders are used ensure that the ladders are in rigid condition, not slip and 1 m above from the top surface for safe and easy access in open cut area having steps by maximum 1.20m height and 1m step.
- Request for Approval (RFA) shall be submitted by attaching all the required documents prior to starting any further activities.

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- Ensure the proper shoring by trench sheet with enough support not to collapse the trench where the loose soil is observed and collapse from rain or dewatering failure.
- If the excavation near to traffic roads that should be barricaded 2.0m away or depends on soil condition by concrete solid barrier.



4.3.6 Storage

- All the material will be delivered to site and will be offloaded in the designated storage area.
- The material will be shifted to the working area using a hi-ab truck or trailer.
- Where hi-ab or trailer not accessible to work site, tele handler will be used, the access arrangements shall be checked by the HSE team before allowing any vehicle or equipment to the work site.

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- GRP and UPVC pipes and fittings will always be stored under cover and out of direct sunlight.
- Material handling and storage will be as per manufacturer's recommendation.
- Pipes of different sizes will be stacked separately.
- The pipes will be stacked on bearers, which must be a minimum of 100 mm wide by 50 mm thick and 3.0-meter apart with adequate wooden checks to prevent movement.
- Pipes and fitting shall be supplied either on pallets or remain on the pallets until required at site.
- UPVC and GRP Pipes (GRP: upto 500mm Nominal Dia) are to be stacked to a maximum number of 5 layers.
- GRP pipes larger than 500mm to 700mm ND shall be stacked to a maximum of 3 layers.
- Pipes shall not rest of their sockets.
- The stacks will be in layers and height based on the dia. of the pipes with a maximum height of 1.5 meter or as recommended by the manufacturer.
- All Pipes and Fittings will be thoroughly inspected and approved by consultants on arrival at site.

4.3.7 Transportation

- Vehicles with flat beds in order to suitably accommodate the full length of the pipes will be used for transporting the pipes.
- Pipes shall be loaded and unloaded at site with the help of Hi-ab / telehandler / Crane.
- Safety approved webbing belts shall be used for loading and unloading of pipes.
- Hiab crane truck shall have EIAC approved certification for the operator and crane.

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4.3.8 Preparation of Formation

- The formation shall be prepared and any stones, pebbles, aggregates etc. shall be removed.
- Formation shall be compacted with plate compactor and tested to ensure 95% of MDD by Field Density Test.
- After the inspection and approval of the formation, laying of geotextile fabric will commence.
- Proper access and egress will be provided to the trench as per standard safety requirements.

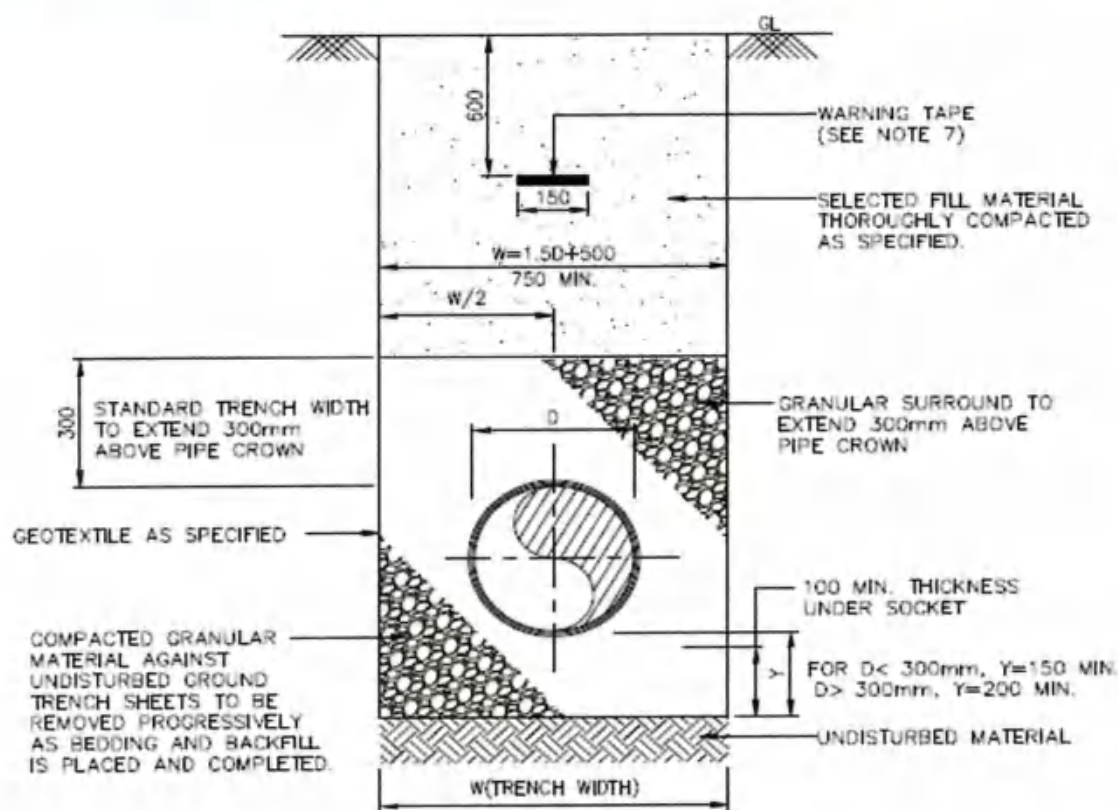
4.3.9 Laying of Geotextile Fabric

- The approved geotextile fabric shall be laid into the trench.
- During the laying of geotextile fabric, the operatives will be protected adequately from accidental falls into the excavation.
- Geotextile fabric shall cover the sides, bottom and top with proper (minimum 500mm) overlaps.

4.3.10 Laying of Granular Material for Pipe Bedding & Surround

- Approved granular material shall be used as pipe bedding and pipe surrounds.
- This granular material shall be laid and spread uniformly.
- The granular bedding below the pipe for dia. < 300mm will be 150mm & dia. > 300mm will be 200mm.
- The granular surround will be laid 300mm above the crown of the pipe in 2 layers of 150mm each.
- When the bedding is at the correct level, and approved by the consultant, laying of the pipe shall be commenced.
- Bedding and surroundings will be compacted manually.

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TYPICAL PIPE BEDDING DETAILS

NOTE:

1. THE TRENCH WIDTHS STATED ARE FOR FULLY STABLE SOIL CONDITIONS.
2. TRENCH WIDTHS IN EXCESS OF THE NOMINAL WIDTHS MAY BE INSTRUCTED WHERE THE EXISTING GROUND IS PROVEN TO HAVE LESS THAN THE REQUIRED PASSIVE RESISTANCE.

4.3.11 Laying of Pipe

- Approved GRP and UPVC Pipes as per approved shop drawings shall be installed.
- The pipes shall be brought to job site and laid along the sides of the trench. Care should be taken to avoid impact on any solid object.
- Safety inspected/approved lifting belts shall be used for shifting and lowered in the trench.
- GRP pipe (Length 12m or less as per site requirement) will be lifted by webbing belts with 2 Legs slings. The angle between the legs of the sling should not exceed 90 degrees. The best angle is 60-degree, and then the 45-degree angle.

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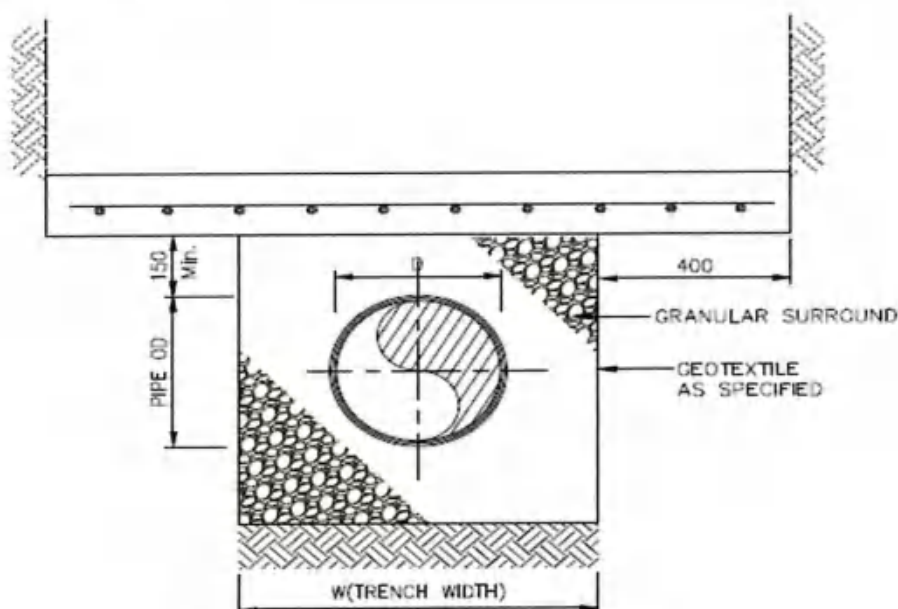
- Lifting equipment will be mobile crane or excavator with approved lifting hook.
- The lifting equipment and belts will be inspected and approved by HSE prior to lifting operation.
- The UPVC pipes will be lifted and laid in the trench either manually or by using webbing belts depending upon depth of trench and site conditions.
- The ends and the inside of the pipe and socket must be clean and free from grit and dust.
- The bedding will be removed from the pipe joints, and pipes should not rest on the joints, and it will be made sure that the bedding provided is firm and continuous over the entire length of pipe.
- Spigots, rubber sealing rings and inside of the socket will be carefully cleaned and examined by the pipe fitter.
- The rubber ring will be inserted and seated in its position inside the socket.
- The chamfered end of the pipe will be lubricated for a min. length of half of the spigot length by lubricant provided by the manufacturer.
- The pipe will be pushed into the socket up to the insertion mark after making sure that the pipes are aligned both horizontally and vertically.
- Pipe will be joined as per manufacturer's recommended procedure.
- Pipe lubricants as supplied by Supplier (as per manufacturer's recommendation) shall be applied to the mating surfaces of the pipes and the sealing ring.
- If pipes require cutting, the cut end should be chamfered to approximately 15° for half the pipe wall thickness and debarred with a sharp-edged tool.
- During joining, both the pipe end and the sealing ring shall be cleaned, lubricated and are correctly aligned.
- The connection to the chambers shall be built with short pipes at both ends of the chamber.
- The length of short pipe shall be as per approved drawings and project specification.

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4.4 Protection of Pipe by Protection Slab and Concrete Surround:

4.4.1 Pre Casted Protection Slab Installation

- The precast protection slabs will be lifted carefully by Hi-AB truck.
- The pre-casted protection slabs will be installed under the supervision of site engineer and a surveyor to achieve the correct orientation, level and coordinate.
- No labour will be allowed to stay in the trench, until the slabs are lowered in the trench.
- Inspection request will be submitted for installation of protection slabs.
- For water proofing 1 coat of primer followed by 2 coats of bituminous emulsion paint (approved material) will be applied for buried concrete as per manufacturer instruction.
- The area shall be backfilled upon completion of all protection works and approval from the Engineer.



PROTECTION SLAB DETAILS

NOTES:

1. PROTECTION SLAB INCLUDED AS SPECIFIED OR INSTRUCTED BY THE ENGINEER.
2. CONCRETE CLASS "A"
3. REINFORCEMENT
USE Ø10 @ 200 C/C IN BOTH DIRECTIONS

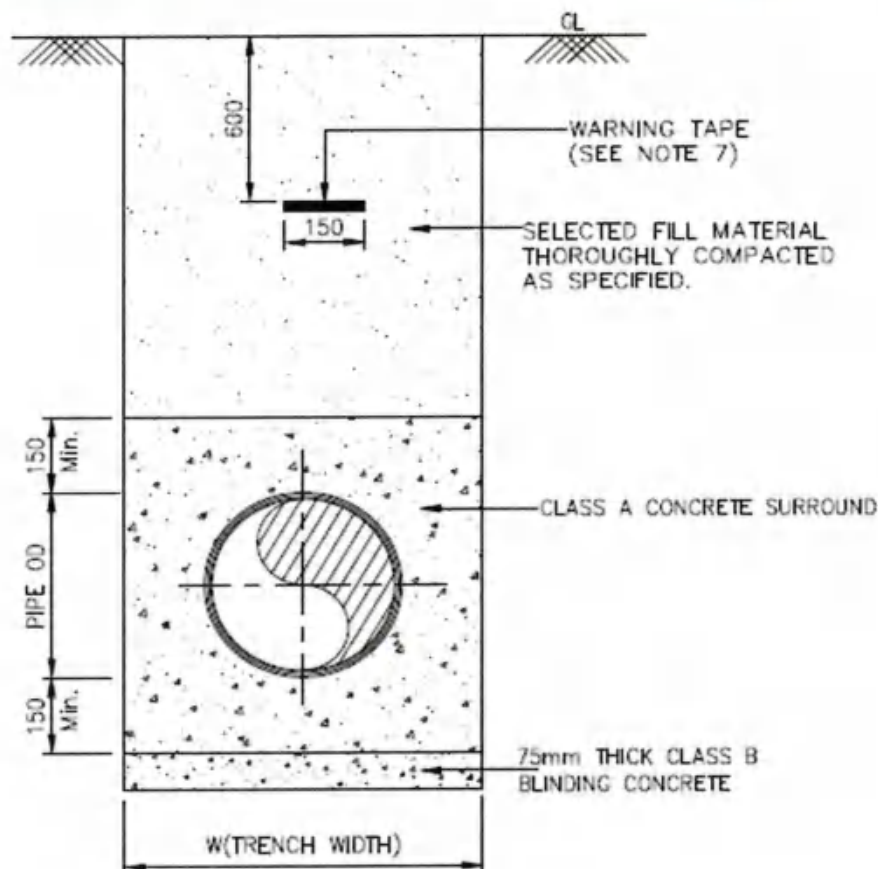
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4.4.2 Concrete Surround

- The concrete surround will be provided in road crossings where depth of pipe is less than 0.90m from F.R.L. or as mentioned on the approved shop drawings.
- Blinding concrete class B of 75mm thickness will be provided below concrete surround as per approved shop drawing.
- Polythene sheets will be provided below blinding concrete to avoid contact of concrete with the soil.
- Approval for blinding concrete will be obtained from consultant.
- Pipe will be placed and aligned over the concrete cradles and formwork will be fixed to the true shape and dimensions as per approved shop drawing.
- Formwork shall have minimum 150mm clearance from the pipes on all sides as per DM Standard.
- Formwork will be cleaned manually from saw dust, loose debris, wire cuttings, wooden pieces etc.
- After inspection and approval of the engineer for formworks, concrete placement will commence.
- Concrete Slump, concrete temperature and cubes will be taken as per relevant project specifications.
- Concrete will be poured as per contract specifications and relevant project specifications.
- Curing (by water upon site conditions) will commence immediately after the free water has left the surface of concrete.
- After curing, 1 coat of primer followed by 2 coats of rubberized bitumen will be applied.
- Inspection and approval by the consultant will be obtained.

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CONCRETE SURROUND DETAILS

4.5 Backfilling

- Suitable material shall be used for backfilling.
- Back-fill material shall be compacted in layers of 150mm by water tamping until the level of back-fill reaches 300mm above the geotextile.
- Further, back-fill material shall be deposited in 200 mm layers and compacted to 95% of M.D.D and tested by large pouring method.
- Back-fill material shall not contain any particles larger than 150mm, debris, waste, vegetation and other deleterious materials.
- Each layer shall be compacted by plate compactor. Any areas inaccessible to the compactors shall be compacted by mechanical tamping.

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- Approved warning tape shall be installed as per the drawing and project specifications.
- Where any vehicle is used for tipping material into any excavation or over the edge of any embankment or earth work, well anchored stop blocks will be used to prevent the vehicles overrunning the edge.
- For each backfill trench, a backfill check sheet mentioning layer's details and pipeline chainage shall be maintained and signed by the consultant at the time of inspection.
- The backfill check sheet will be attached with the backfill inspection request.
- Backfilling must reach until finish ground level.

4.6 Testing & Commissioning

- Testing of Pipeline will be carried out as per the project specifications.
- UPVC pipelines will be subjected to:
 - Preliminary Air Test after granular surroundings,
 - Water test & Mandrel Test after backfilling.
- GRP pipelines will be subjected to a water test after backfilling (to check for the leakage) and a final deflection test (to check for deflection).
- During testing, warning signs and watchman will be assigned to alert and control unauthorized personnel.
- The testing area will be barricaded, and access will be restricted to TC Team only. Continued supervision will be provided.
- The testing and commissioning will be conducted by personnel trained for the task.

4.6.1 Air Test (GRP & uPVC PIPES)

- Before backfilling, pipeline will be given a preliminary air test as detailed below:
- All pipe outlets will be plugged with suitable plugs and braced where needed.
- Air shall be pumped in slowly to the pipe until a pressure of 100mm head of water is indicated on a manometer connected to the system.

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- After the internal pressure of 100 mm is obtained, 2 minutes shall be allowed for the air temperature to stabilize within the pipe.
- Air may be added to restore the pressure to 100 mm.
- The test shall be considered passed, if, during a further period of 5 minutes (without further pumping of air) the pressure should not fall below 75 mm.
- Air pressure will be released by removal stopper at far end of the pipeline away from the manometer.

4.6.2 Water Test (UPVC & GRP) Pipes

- After backfilling of pipeline, water test shall be conducted by as detailed below:
- All branches and open ends shall be closed with suitable stoppers.
- Water shall be filled from the lowest point and air will allow us to escape. No entrapped air shall remain in the pipeline while testing.
- An internal pressure equivalent to a head of water of depth to invert of the pipeline plus 1.25m, or 4 m head whichever is greater shall be maintained for 30 minutes to allow for initial absorption of water.
- After that, the test pressure shall be maintained for 60 minutes and water added shall be measured.
- Pipelines shall be treated as passed if water consumption in 60 minutes does not exceed 7.5L/m per 30m of length of pipeline under test.

4.6.3 Deflection Test by Mandrel (uPVC Pipelines)

- For uPVC pipelines deflections shall be checked by, pulling through it, a cylindrical mandrel suiting the permissible minimum deflected diameter of the pipe.
- The pipeline will be considered passed if the mandrel passes through it.
- For uPVC pipes 3% deflection allowed for initial and 5% of Nominal Diameter for long term.

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4.6.4 Deflection Test by Deflectometer (GRP Lines)

- A deflectometer capable of measuring diametric dimensions both vertically and horizontally for deflection measurement of pipes will be used.
- The deflectometer shall be calibrated for each diameter regularly in the presence of the Engineer to maintain the accuracy of the instrument.
- The device shall provide location for each deflection measurement along the length of pipeline.
- For pipes of 600 mm diameter and greater the device shall be having a telescopic spring-loaded rod graduated for each diameter to show percentage deflections.
- The design of pipe bedding and backfilling method shall be such that the initial and long-term deflection does not exceed the following values as specified in project specification clause PS 6/62.2 sub clause 15:
- Initial - after backfilling: 2% of internal diameter
- months after installation: 4% of internal diameter (if the newly laid GRP pipeline has to be made live before 6 months' time, then only initial deflection test shall be conducted after backfilling).
- Deflection tests shall be conducted for newly laid GRP lines spanning a length more than 300m and for pipe diameter greater than or equal to 600mm only.

4.7 Construction of Manholes & Gullies

- Chambers will be cast in the following stages.
- Construction of Standard Base Slab, Cover Slab & Gullies (at Precast yard):
- Shifting and installation of Base and Gullies to site.
- Casting of Manhole Wall at site.
- Waterproofing, Pipe Connection and Backfilling.
- Shifting & Installation of Cover Slab at site.
- Installation of Cover & Frame and Haunching Concrete.

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4.7.1 Material for Manhole Construction

The materials required are as follows:

- GRP Benchings and Liners 7mm
- Approved Ready Mix Concrete
- Approved Epoxy Coated Steel reinforcement
- Built in pipe short pieces
- Grit Bonding materials
- Gully pot with fittings
- Cover and Frame
- Concrete repair materials
- Waterproofing materials
- Concrete repairing material.
- Suitable backfill Material.
- Scaffolding materials
- Trench sheet H-Beam supports (For shoring if required)
- Plastic Barriers, Concrete Barriers, Cones, Ropes, Warning Tapes & Warning Signs.

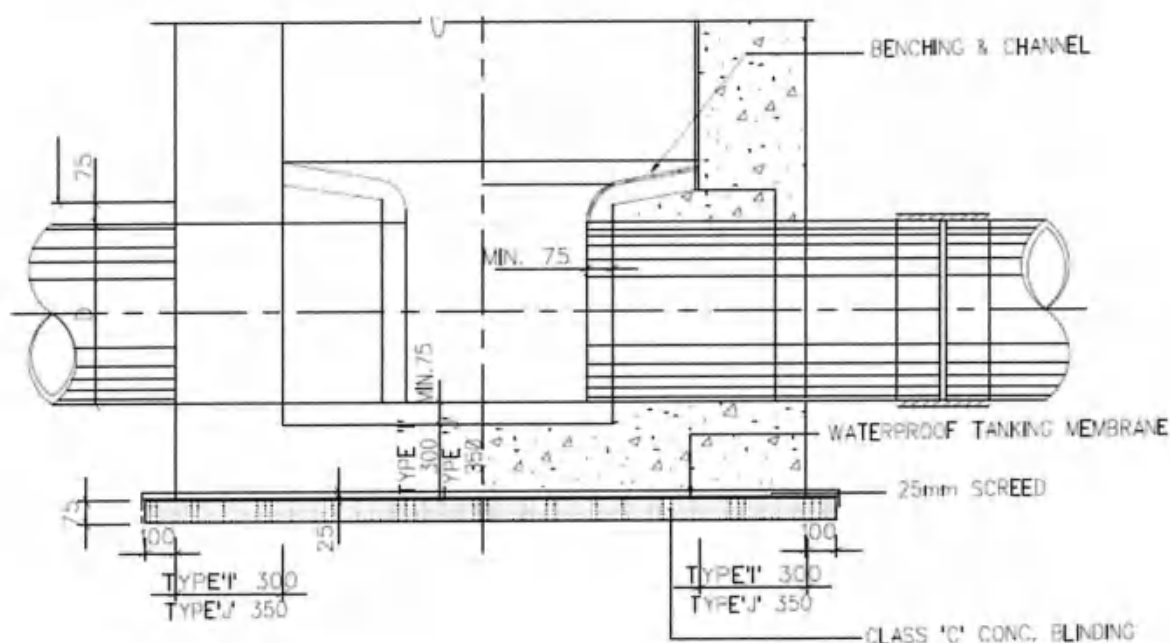
Note: All material sources / suppliers shall be separately submitted and approved prior to use at site.

4.7.2 Construction of Standard Base

- The standard base slab will be casted using a prefabricated approved GRP benching 5mm thick as per approved shop drawing.
- Formworks true to the shape and dimensions will be fixed for base slab.
- Approved reinforcement, lifting hooks and short pieces with grit bonding will be fixed as per approved shop drawing.
- Reinforcement steel bars shall be free from oil, grease, tar, rust or other deleterious material prior to fixing.
- Concrete spacers will be used to provide sufficient cover for steel reinforcement.

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- On completion of fixing rebar, the area Inside the form will be cleaned from debris, bits of binding wire and dust.
- Base slab will be inspected & approved by consultant prior to commencement of concrete.
- Form releasing agent (oil) will be applied to formworks prior to concrete casting.
- Approved concrete mix will be used.
- Concrete Slump, concrete temperature and cubes will be taken as per relevant project specifications.
- Concrete will be placed by means of Mobile Crane / concrete skip / directly poured from concrete mixer truck for low height falling (less than 2m).
- The concrete will be poured as per contract specifications and relevant project specifications.
- Curing (by water or curing compound depending on site conditions) will commence immediately after the free water has left the surface of concrete.
- Inspection and approval by consultant will be obtained.



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4.7.3 Construction of Standard Cover Slab at Pre-Cast Yard

- The standard cover slab will be casted by fixing inner and formwork as per approved shop drawing dimensions.
- Approved Reinforcement steel (free from oil, grease, tar, rust or other deleterious material), lifting hooks and concrete spacers will be fixed as per approved shop drawing.
- After approval of formwork and steel reinforcement, concrete will be poured as per relevant project specifications.
- Inspection and approval from consultant will be obtained.

4.7.4 Construction of Standard Gully at Pre-Cast Yard

- The standard gully will be casted as per approved shop drawing.
- Formworks true to the shape and dimensions will be fixed for base.
- Approved gully pot, lifting hooks and gully adaptor with grit bonding will be fixed as per approved shop drawing.
- On completion of fixing formwork, the area inside the form will be cleaned from debris, bits of blinding wire and dust.
- Inspection and approval from consultant will be obtained prior to commencement of concrete.
- Form releasing agent (oil) will be applied to formworks prior to concrete casting.
- Approved concrete mix will be used.
- Concrete Slump, concrete temperature and cubes will be taken as per relevant project specifications.
- Concrete will be placed by means of Mobile Crane / concrete skip / directly poured from concrete mixer truck for low height falling (less than 2m).
- The concrete will be poured as per contract specifications and relevant project specifications.

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- Curing (by water or curing compound depending on site conditions) will commence immediately after the free water has left the surface of concrete.
- Inspection and approval by consultant will be obtained.

4.7.5 Excavation & Formation Preparation for Manhole

- Setting out will be carried out as per the approved shop drawing.
- The excavation will be done by open cut method and formation level will be watered and compacted to obtain 95% of M.D.D. as detailed in the "excavation and formation preparation" section of pipe line of this document.
- Work inspection request (WIR) will be submitted to consultant for the approval of formation.

4.7.6 Blinding Concrete, Waterproofing, Shifting & Installation of Base

- The setting out for chamber blinding concrete of 75mm thickness according to the center point will be marked by surveyor.
- Polyethylene sheets will be provided to avoid contact of blinding concrete with the soil.
- Approved blinding concrete will be placed.
- Water proofing membrane and protection board/screed will be provided on blinding concrete as per manufacturer's recommendations.
- Request for Approval (RFA) will be submitted to consultant for the approval of blinding, waterproofing and screed.
- Manhole Base will be shifted and laid down in the exact location using a mobile crane/hi-ab with flatbed trailer and safety precautions will be taken as stipulated in attached risk assessment.
- The base will be installed under the supervision of site engineer and surveyor to achieve the correct orientation, level and location.
- Inspection and approval for base installation will be obtained from consultant prior to proceeding to next activity.

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4.7.7 Shifting & Installation of Gully at Site

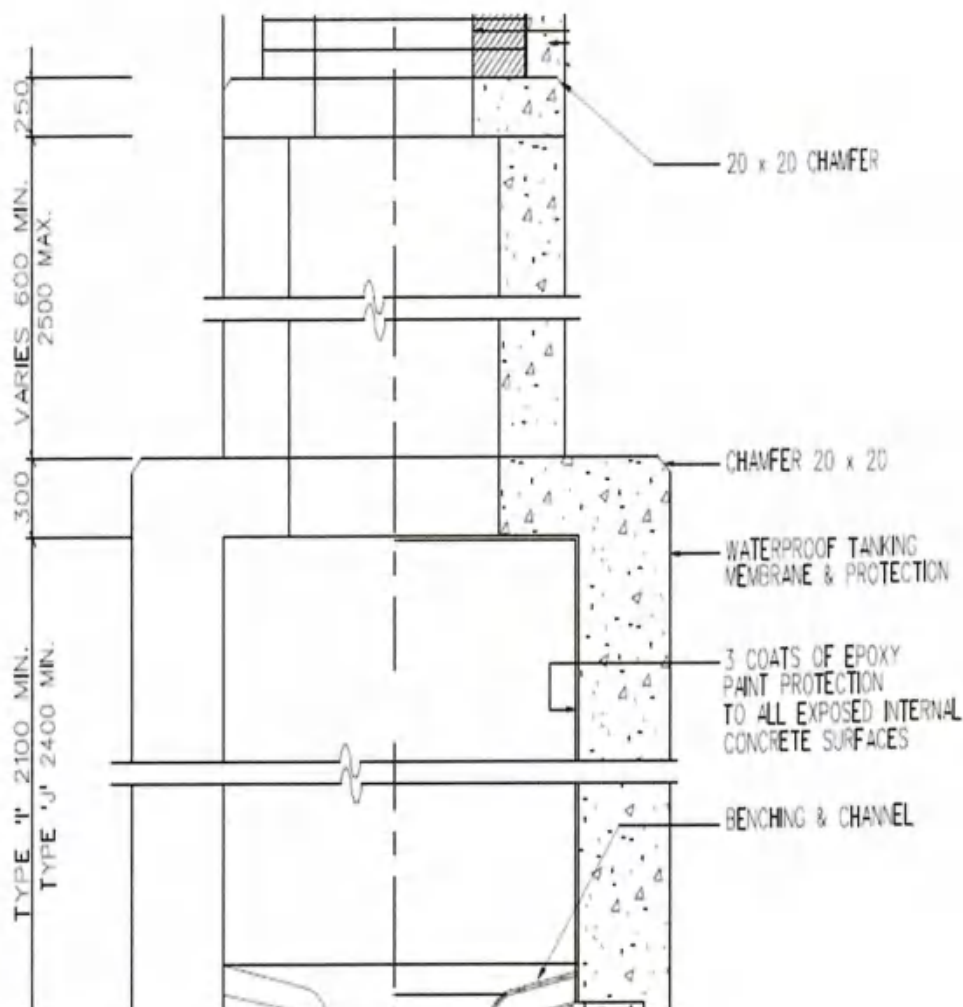
- After the approval of formation, geotextile will be provided and gully will be shifted and laid down in the exact location using a mobile crane/hi-ab and safety precautions will be taken as stipulated in attached risk assessment.
- The gully will be installed under the supervision of site engineer and a surveyor to achieve the correct orientation, level and coordinate.
- No labor will be allowed to stay in the trench, until the slabs are lowered in the trench.
- Inspection request will be submitted for installation of protection slabs.
- For water proofing 1 coat of primer followed by 2 coats of bituminous emulsion paint (approved material) will be applied for buried concrete as per manufacturer instruction.
- The area shall be backfilled layer by layer as per standard DM Procedure as mentioned above in pipe section upon completion of all connection work and approval from the Engineer.

4.7.8 Construction Of Manhole Wall at Site

- Once the base is installed at site, the prefabricated 5mm GRP Liners will be used as internal shutters, supported from inside by wooden shutters and props. Water stoppers will be fixed at each construction joint.
- Outside wooden/GRP shutters will be erected and fixed.
- Scaffolding will be erected outside for access and concrete casting.
- Only essential tools, materials and personnel on work will be allowed to go over the platform.
- The formwork stability must be ensured during positioning operations and structural support props should be fixed and adjusted to ensure correct line and verticality is achieved. All through ties and props must be correctly positioned and at the right centers.

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- On completion of fixing formwork, the area inside will be cleaned from debris, bits of binding wire and dust.
- Formwork will be inspected & approved by consultant prior to commencement concrete.
- Form releasing agent (oil) will be applied to formworks prior to concrete casting.
- Concrete will be cast as per the relevant project specifications.
- Curing (by water or curing compound depending on site conditions) will commence immediately after the free water has left the surface of concrete.
- Inspection and approval by consultant will be obtained.



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4.7.9 Waterproofing

- Approved waterproofing system will be used in accordance to manufacturer's recommendation and guidelines.
- For external surfaces water proofing membrane with protection board will be used.
- The waterproofing system will be inspected and approved by the engineer.

4.7.10 Pipe Connection to Manhole

- Pipe connection to manhole will be carried out upon completion of water proofing.
- Coupler joints will be utilized for flexible joints.
- Protection slabs will be provided to the flexible joints if required by the engineer as per site conditions.
- Connection to existing chamber or pipeline line shall be coordinated with concerned authorities by providing plan, program, time bar with sequence of work, shutdown procedure etc.

4.7.11 Backfilling

- Backfilling with suitable material will be done layer by layer as detailed in the backfilling section for pipeline as per DM Standard procedure, and approval from consultant will be obtained.

4.7.12 Shifting & Installation of Cover Slab

- The standard cover slab will be shifted and placed in the exact location with a mobile crane.

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4.7.13 Installation Of Cover, Frame & Haunching Concrete

- Upon completion of backfilling, cover frame will be installed on the top of cover slab as per approved shop drawing. It may be fixed on the cover slab or fixed over precast rings as per approved shop drawing.
- Cover & frame size shall be 600mm opening diameter.
- Various layers of precast extension rings (not less than 2 not more than 5) with will be laid with cement mortar to adjust the finishing level of the chamber cover.
- On the top of precast extension rings cement mortar will be laid by means of a trowel and frame will be installed on this mortar as per the finish levels.
- Upon completion of frame installation square shaped plywood shutter will be fixed around the frame reaching up to top level of frame 20 x 20mm PVC chamfers will be fixed on the corners of the formwork to avoid any sharp edges on the haunching concrete.
- Concrete pouring and curing will be carried out with the same procedure as mentioned above.
- Smooth surface will be achieved by means of a trowel on the top of haunching concrete.
- Approval from consultants will be obtained.
- The top cover level shall be flushed with the F.R.L.

4.7.14 In-Situ Lamination Works

- All the internal joints and UPVC pipes connecting points to the shaft liner will be in-situ laminated.
- The lamination will be done using glass chopped strand mats impregnated with isophthalic resin and vinyl ester resin.
- Firstly, the surface receiving lamination shall be coated with isophthalic resin.
- Three layers of 600 gm/m² glass chopped strand mat shall then be applied and impregnated with isophthalic resin to provide an approximate thickness of 5 mm.

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- Next one layer of 600gm/m² and one layer of 300gm/m² glass chopped strand mat shall be applied and impregnated with vinyl ester resin.
- The following two layers of glass veil shall be applied and impregnated with vinyl ester resin.
- Finally, a coating of vinyl ester resin with paraffin wax shall be applied to provide a total minimum thickness of 5mm.

5. Activity Hold Points

As reference for HSE, general hold activities related to work are:

- Permit to work
- Risk Assessment
- Induction and Pre-start completed
- Plant and equipment safety checks completed

6. Inspection and Records

All inspection / check list and document records shall be based on field notes taken during activities and shall be recorded

HSE records shall include:

- Toolbox Talk Record
- Equipment inspection check-sheet (As applicable)
- Competency of operators, flagman and all site team involved in above mentioned method statement to be verified by the CCECC.

7. Safety and Environmental Risks

Hazards and risk assessment are carried out and attached in this Method Statement in the form of a Risk Assessment. These are reviewed by the HSE Manager / HSE Officer to ensure that all the hazards have been identified and adequate control measures are in place.

All personnel should wear the appropriate PPE.

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7.1 Hazard & Risk Assessment

A Risk Assessment will be completed by the section performing the work together with the HSE Department. This document will identify hazards and develop strategies to mitigate the risks. All personnel carrying out the work will be competent and experienced. Where applicable, certificates of competence shall be provided.

Refer to Attachment-A.

7.2 Tools and Equipment

All tools and equipment for the task shall be in a safe condition and fit for its purpose. Where applicable, registration papers and certificates are to be provided.

At the start of each shift, equipment shall be visually checked for mechanical and structural soundness. All equipment shall be carefully examined and all hoses and couplings checked for leaks and general conditions.

7.3 Accident / Incident

In the event of an incident, response will be as per the procedure. Prestart and Toolbox meetings will be carried out to reinforce safety requirements prior to commencement of the works.

7.4 Protective and Safety Equipment

Mandatory Personnel protective equipment to be worn by all staff and operatives on this site to carry out the tasks of this method statement are:

- Safety Helmet
- Safety Boots
- High Visibility vest
- Safety goggles
- Safety Gloves as and when required
- Dust mask as and when required
- Ear protection as and when required

7.5 Welfare Arrangements

- All Activities will comply with Dubai Municipality requirements.

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7.6 Access / Egress to Construction Area

- Clearly marked and designated access/egress routes should be planned. These paths should lead to / from the ground level in a manner that minimizes hazards and is easily recognizable.
- Routes should be free from debris or any other obstacles that could impede movement. Regular inspections should be in place to ensure these routes remain clear.
- Clear signage indicating hazardous areas, such as edge protection, open holes, or overhead work, should be placed near access points to alert workers.
- Workers should be trained on the correct methods of accessing and egressing from the work area
- Access Constructions Vehicles Only' and flagman for guiding the vehicles into construction area.
- Prevent taking rest & sleeping in the workplace.
- Driver / Operator to check the persons below the plant / equipment before starting.
- Mobile Crane with Man Basket will be available at site.
- Provide deceleration lane close to the speed / live track & provide a sign board.
- Appropriate Signs and traffic barrier as per approved traffic management plan.
- Access Constructions Vehicles Only' and flagman for guiding the vehicles into construction area.
- Minimum speed limit to be strictly followed.
- Heavy equipment check list to be followed before start of activity.
- Prevent taking rest & sleeping underneath the plant.
- Driver / Operator to check the persons below the plant / equipment before starting.
- Employees working near live traffic are ensure not cross on the live traffic, standby vehicle will transport them on the other side of the road.

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7.7 List of Training

1	Emergency response	
2	PPE	
3	Housekeeping and waste segregation	
4	Lifting operations	
5	Power tools	
6	Electrical safety	
7	Heat stress	
8	Manual handling	
9	First Aid	
10	Chemical Storage and Handling	
11	Traffic Management	
12	Banksman	
13	Environmental Programs	
14	Excavation	
15	Fire safety	
16	Materials storage safety	
17	Spill Training	
18	Rigging	
19	Lifting Operations, Slings	
20	Noise & DUST Control	

Note: The required training shall be provided at the time of site activity, based on the specific task, risk level, and site conditions.

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7.8 First Aid

- A fully stocked first aid kit shall be present on site at all times, accessible and regularly inspected.
- A minimum of one certified First Aider shall be present during all work activities.
- Emergency transportation shall be available in case medical evacuation is needed.
- A designated First Aid Station shall be established if the site conditions and manpower size require.

7.9 Lighting

- Good light will be provided— use natural light where possible and avoided glare
- A good level of local lighting at Workstation where necessary will provided.
- Suitable forms of emergency lighting
- Well-it corridors and Access.
- Well-it outside areas – for pedestrians and to help with work activities such as loading/unloading at night.

7.10 Working Outdoors

- For work outdoors we consider things such as the weather, temperature (both hot and cold) and sun exposure. Work will be as per DM CODE OF SAFETY & APPROVED SUMMER PLAN.

7.11 Moving Around the Premises

- Safe passage for pedestrians and vehicles – separate routes will be provided.
- Personal Protective Equipment is mandatory
- Levelling, even ground surfaces without holes or broken boards
- Handrails and ramps where necessary
- Set up properly secured fall arrest systems (like lanyards connected to fixed anchor points).
- Ensure no work is carried out during adverse weather conditions like high winds or rain, which increase the risk of falls.
- Create designated areas for safe movement and keep others away from high-risk zones.
- Safely constructed Access.
- Surfaces which are not slippery.

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- Earth Moving Activity will be ensured as per the approved Earth Moving Equipment Plan.

7.12 Emergency Response Procedures

"Emergency response procedures shall be implemented in accordance with the approved Project Emergency Preparedness and Response Plan (PEPRP), Doc. Ref.: CCECC-R1181/2-PEPR-PLN-07. In addition, activity-specific emergency arrangements shall be established considering Storm water works including excavation, live traffic interface, dewatering, lifting operations, Confined Space and pipeline testing."

Emergency scenarios shall include:

- Excavation/trench collapse
- Utility strike (DEWA / telecom / drainage)
- Traffic accident within work zone
- Flooding due to dewatering failure or Pressurized pipeline failure during testing
- Fire or fuel spill
- Medical emergency / heat stress
- Confined Space

In case of emergency:

- Work shall be stopped immediately; Area shall be secured and isolated
- Emergency alarm shall be raised.
- Supervisor shall take control as Incident Controller
- Emergency services shall be contacted immediately
- Emergency assembly points shall be clearly marked and communicated to all personnel. Muster control shall be implemented to ensure all personnel are accounted for.
- Emergency contact details (Ambulance, Civil Defense, Police, nearest hospital, project emergency team) shall be displayed at site.
- Access routes for emergency vehicles shall be maintained clear at all times.

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- Communication system (radio/mobile) shall be available to ensure immediate coordination during emergency.
- First aid facilities and trained first aiders shall be available at all times.
- Emergency drills shall be conducted periodically for: Evacuation, Excavation, collapse, Traffic incident, Flooding scenario, Confined Space

All personnel shall be trained on emergency response protocols during the initial site induction.

7.13 Rescue Arrangements

Rescue arrangements shall be specific to irrigation construction activities and shall not rely on generic or unrelated methods.

Excavation / Trench Rescue:

- No entry into collapsed trench without assessment
- Trained rescue team shall respond
- Use of ladders, shoring support, and safe access
- Emergency services to be contacted immediately for major collapse

Confined Space (if applicable – manholes):

- Tripod, harness, lifeline system shall be used
- Standby man present at all times
- Gas monitoring and ventilation ensured

Flooding / Dewatering Failure:

- Immediate evacuation from excavation
- Pumps activated / backup system engaged
- Area secured to prevent re-entry

Traffic-related Incident:

- Area secured using traffic control measures
- Casualty moved only if safe
- Coordination with Traffic Police and ambulance

Lifting Operation Incident:

- Exclusion zone maintained
- No attempt to move suspended load manually
- Crane isolated and area secured

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Medical Emergency / Heat Stress:

- Immediate first aid, Transfer to shaded/cooling area
- Ambulance called if required

Rescue Equipment Availability:

- Stretchers, first aid kits, Rescue ropes, harness
- Spill kits, Fire extinguishers,

Rescue Team & Responsibility:

- Trained personnel assigned for emergency response
- Roles clearly defined and communicated

All rescue arrangements shall be briefed during toolbox talks and verified through drills.

Mock Drills: Regular emergency mock drills shall be conducted to ensure all personnel are familiar with rescue procedures and equipment use.

7.14 Emergency Contact Details

In case of Emergency Contact Telephones, No.:

- Police: 999
- Ambulance: 998
- Civil Defence: 997
- DEWA: 991
- HSE Coordinator: 0561487012
- HSE Manager: 0559331082

8. Appendices

8.1 Appendix A: Risk Assessment

8.2 Appendix B: Inspection & Test Plan

8.3 Appendix C: Checklist

8.4 Appendix D: Emergency Contact Details

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Appendix A: Risk Assessment

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HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

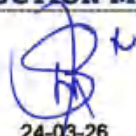
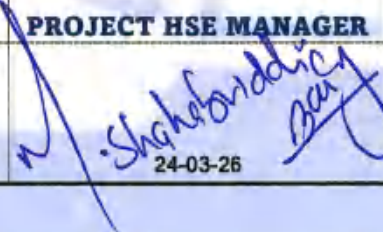
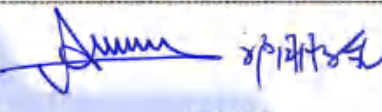
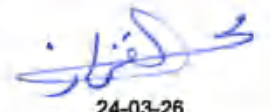
RISK ASSESSMENT FOR DM - STORM WATER NETWORK

PROJECT: R1181-2 - IMPROVEMENT OF AL SAFA ROAD

1. DOCUMENT CONTROL.

Contractor / Subcontractor:	China Civil Engineering Construction Corporation - Dubai Branch	Project Location:	Al Safa Road (Financial Cent Road to Al Wasl Road)	Last Revision Date:	Not Applicable
Method Statement Ref. No:	CCECC-R1181-2-CC-UT-MSRA-022	Risk Assessment Ref. No:	CCECC-R1181-2-HSE-RA-22	Next Review Date:	15-10-2026
Assessment Date:	24-03-2026	Activity Type:	Routine ☒ Non-Routine ☐	Revision No:	00

This Risk Assessment shall be reviewed and updated upon any change in methodology, design, equipment, regulatory requirements, traffic stage/diversion, identification of utility conflicts, or following any accident or incident.

REVIEW & AUTHORIZATION	RISK ASSESSED BY ENGINEER + SITE MANAGER +CM	REVIEWED BY HSEM	REVIEWED BY DPM	AUTHORIZED BY PD
NAME:	AHMAD GHANEM	MIRZA SHAHABUDDIN BAIG	AHMED ELKADI / GUO PENGCHENG	MOHSEN D GAZZAZ
DESIGNATION:	CONSTRUCTION MANAGER	PROJECT HSE MANAGER	DEPUTY PROJECT MANAGER	PROJECT DIRECTOR
SIGNATURE:				
DATE:	24-03-26	24-03-26	24-03-26	24-03-26

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR DEWA- WD STORM WATERNETWORK

2. SCOPE OF WORK:



- Execution of Storm Water Works including protection of existing services, access arrangements, setting out, dewatering, excavation & formation preparation, GRP pipe laying (storage, bedding, geotextile, surround, protection slab & thrust blocks), backfilling, road crossing ducts & testing, chamber construction & installation, valves, waterproofing, pipe connections, cover slab & haunching concrete, pressure & deflection testing, and all related activities as per Approved Shop Drawings, Project Specifications and Method Statement.

3. SUMMARY OF PLANT & EQUIPMENT, MATERIALS, TOOLS



- Plant:** - Excavator / JCB, Mobile Crane / Hi-Ab with flatbed trailer, Plate Compactor, Concrete Mixer Truck/Skip, Dewatering Pumps, Total Station/Auto Level, Mandrill Test Kit, Lifting Gear (slings, shackles, chains), Tower light.
- Tools:** - Hand Tools, Vibrators, Power Tools (Hand Grinder, Circular Saw, Hand Grill etc.)
- Materials:** - GRP Pipes & Bedding Material, Geotextile Fabric, Ready-Mix Concrete, Polythene Sheet, Rubberized Bitumen, Precast Slabs / Rings/ Extension Rings, Valves & Fittings, Waterproofing Membrane & Epoxy Paint, Warning Tape.

4. PERMITS, NOCS



- Internal Permits:** Excavation, Lifting, Confined Space, Hot Work, Electrical PTW and Work at Height (if required).
- Authority NOCs:** RTA ROW Permit, DEWA Utility NOC, DU/Etisalat NOC (if applicable).

5. SUMMARY OF SPECIFIC TRAINING (INTERNAL & EXTERNAL)



- Internal Training** HSE Induction, MSRA, PTW Awareness,
- External Training / Certified:** Crane Operator, Rigger / Banksman, Lifting Supervisor, First Aid & Fire Fighting, Heavy Equipment Operator.

6. SUMMARY OF PERSONAL PROTECTIVE EQUIPMENTS



- Mandatory:** Helmet, High-Visibility Vest, Safety Shoes,
- Task-Specific:** Hearing Protection, Gloves, Goggles, Face Shield, Dust Mask, Full Body Harness (if required).

7. LEGAL REFERENCE / REQUIREMENTS



- UAE Federal Laws (Labour & Environment),
- DM Local Order 61/1991 (CoCSP),
- RTA QHSE MSTG
- RTA ROW & WZTM Manual,
- RTA ROW & WZTM Manual,
- DEWA Technical Regulations,
- EIAC Lifting Certification,
- Dubai Civil Defense Code.

8. SUMMARY OF INSPECTIONS



- Daily HSE Inspection, Excavation/Bore Stability Check, Working Platform Verification, Plant Pre-Start Inspection, Crane & Lifting Gear Certification Check, Electrical DB Inspection, Pile Integrity/Load Test Inspection, Traffic Diversion Inspection

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR DEWA- WD STORM WATER NETWORK

9. RISK ASSESSMENT MATRIX -

A. OH&S RISK RATING CATEGORIES & ACTION LEVEL ↓			B. DURATION OF RISK EXPOSURE & AREA OF INFLUENCE ↓			C. RISK MATRIX INTERACTION (L × S) 5×6 MATRIX ↓ (HEAT MEAP)								
RISK LEVEL	SCORE RANGE	REQUIRED ACTION	RISK LEVEL	DURATION	AREA	SEVERITY 5 Level, Vertical	LIKELIHOOD - 6 levels, horizontal							
L - Low ACCEPTABLE	1-4	Before Task: Additional Controls Commences + Management Approvals Required	LOW L	SHORT TERM (< 1 Month)	LIMITED (WITHIN SITE BOUNDARY)				1- REMOTE	2- UNLIKELY	3- RARE	4- LIKELY	5- COMMON	6-ALMOST CERTAIN
M - Medium Moderate	5-9	Basic controls before task starts	MEDIUM M	MEDIUM TERM (< 1 MONTH)	LIMITED (WITHIN SITE BOUNDARY)		5 Catastrophic	HIGH	HIGH	VERY HIGH	VERY HIGH	VERY HIGH	VERY HIGH	VERY HIGH
H - High Unacceptable	10-15	Before Task: Additional Controls Commences + Management Approvals Required	HIGH H	LONG TERM (1-6 months)	EXTENSIVE (OFF-SITE Community)		4 Major	MEDIUM	MEDIUM	HIGH	HIGH	VERY HIGH	VERY HIGH	VERY HIGH
Very High PROHIBITED	16-20	Immediate Stop - No Work	VERY HIGH V	VERY LONG TERM (> 6 months)	WIDESPREAD (OFF-SITE COMMUNITY)		3 Moderate	MEDIUM	MEDIUM	HIGH	HIGH	HIGH	HIGH	HIGH
							2 Minor	LOW	LOW	MEDIUM	MEDIUM	HIGH	HIGH	HIGH
						1 Incidental	LOW	LOW	LOW	MEDIUM	MEDIUM	MEDIUM	MEDIUM	
D. SEVERITY SCALE (1 - 5) ↓			E. LIKELIHOOD SCALE (1 - 6) ↓											
SEVERITY ↓	MEANING	ENVIRONMENT IMPACT	SAFETY IMPACT	HEALTH IMPACT	LIKELIHOOD ↓	MEANING	PRACTICAL DESCRIPTION	CONSTRUCTION CONTEXT						
5 CATASTROPHIC	Multiple or single fatality / major system failure	Major environmental damage, regulatory shutdown, widespread impact	Fatality or multiple fatalities	Life-threatening irreversible injury	1 REMOTE	Extremely unlikely	No known occurrence in similar controlled works	Fully engineered excavation with certified shoring						
4 MAJOR	Permanent disability or severe injury	Significant environmental impact requiring authority intervention	Permanent disability amputation	Serious medical condition requiring hospitalization	2 UNLIKELY	Possible but not expected	It has occurred in industry but rare under controls	Utility strike despite trial pits						
3 MODERATE	Medical treatment case / Lost Time Injury	Noticeable environmental impact requiring corrective action	Minor injury requiring first aid	Temporary illness or recoverable injury	3 RARE	Occasional occurrence	It has occurred in similar projects	Minor trench instability						
2 MINOR	First aid case	Localized minor environmental impact	Minor injury requiring first aid	Minor reversible health effect	4 LIKELY	Expected to occur sometimes	Occurs periodically if controls not strict	Expected to occur sometimes						
1 INCIDENTAL	No injury / negligible consequence	Negligible environmental impact	No injury	No health effect	5 COMMON	Occurs frequently	Repeated exposure to routine tasks	Dust exposure without suppression						
					6 ALMOST CERTAIN	Will occur without control	Happens regularly in absence of controls	Slip / trip without housekeeping						

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR DEWA- WD STORM WATER NETWORK

DATE: 23-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
1.	Site Mobilization - Storm Water Works	Deployment of unfit/untrained personnel; improper setup of temporary facilities; Uncontrolled movement of plants and vehicles; lack of access control; unsafe material unloading; interaction with live traffic; utility damage; defective equipment; adverse weather conditions	Personal injury. Vehicle collision, property damage, utility strike, public impact, regulatory penalties, project delay	3 Likely	4 Major	12 High	Site mobilization shall be carried out only after obtaining all required approvals, permits, and authority clearances including valid RTA ROW permit and access permissions. Mobilization activities shall be planned in advance and executed in accordance with approved logistics and mobilization plan. All personnel shall undergo mandatory HSE induction prior to site entry covering site hazards, emergency procedures, traffic risks, and reporting protocols. Verification of Emirates ID, valid visa, and Workmen Compensation Insurance shall be completed prior to deployment. Pre-employment medical fitness shall be ensured for all workers prior to mobilization. Welfare arrangements including drinking water, shaded rest areas, and first aid facilities shall be established prior to commencement of site activities. Compliance with Dubai Municipality Summer Working Hours regulation shall be strictly maintained. Temporary site facilities including offices, storage areas, welfare facilities, and utilities shall be installed on stable ground and in approved locations. Electrical and water connections shall be carried	Conduct periodic welfare inspections and health monitoring. Re-inspect scaffolds, excavations and temporary structures after adverse weather before resuming work. Suspend work during severe weather conditions including lightning, sandstorm or high winds. Increase random document verification inspections and conduct daily pre-start briefings.	1 Remote	4 Major	4 Low	HSEO / Admin / Supervisor / Male Nurse

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR DEWA- WD STORM WATERNETWORK											DATE: 23-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							out by authorized personnel and inspected prior to use. All plants, equipment, and materials shall be mobilized through designated access routes in coordination with Traffic Management Team to ensure safe movement and avoid disruption to public traffic. Delivery schedules should be controlled to prevent congestion and unsafe conditions. Material unloading shall be carried out using appropriate lifting equipment under supervision, and exclusion zones shall be established to prevent unauthorized access during unloading activities. Site boundaries shall be clearly defined and secured using fencing, barriers, and signage to control access and prevent entry of unauthorized personnel. Security arrangements should be implemented where required. Lone work shall be prohibited during mobilization activities, and new workers shall be paired with experienced personnel through a buddy system. All plants and equipment shall have valid third-party certification and shall be inspected prior to use. No equipment shall be allowed to					

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR DEWA- WD STORM WATERNETWORK

DATE: 23-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							operate without HSE clearance and inspection tagging. Existing underground and overhead utilities shall be identified, verified, and protected prior to installation of temporary facilities or movement of plant. Weather conditions shall be monitored through official forecasts, and mobilization activities shall be adjusted accordingly. Adequate lighting shall be provided during low visibility conditions. Good housekeeping shall be maintained to ensure organized site layout and safe movement of personnel and equipment. Pre-task briefings shall be conducted to communicate mobilization sequence, hazards, and control measures. Continuous supervision shall be ensured by Site Engineer, Logistics Team, and HSE personnel to monitor compliance and immediately correct unsafe practices. Mandatory PPE including helmet, high-visibility vest, safety shoes, and gloves shall be worn at all times. Additional PPE should be used as required.					
2.	Traffic Management Setup	Incorrect or unsafe diversion installation; hit by moving vehicles during setup/removal;	Fatality / Multiple casualties Serious injury Traffic disruption Property damage	4 Likely	5 Catastrophic	20 Very High	Traffic management setup shall not commence without approved Traffic Management Plan (TMP), valid RTA ROW permit, and all related	Deploy TMA during setup, shifting, and removal of traffic control devices in live traffic interface areas.	1 Remote	5 Catastrophic	5 Medium	TTM Manager / Traffic Engineer /

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RISK ASSESSMENT FOR DEWA- WD STORM WATERNETWORK											DATE: 23-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
		inadequate advance warning signage; improper barrier alignment; insufficient buffer zone; poor night visibility; premature lane closure; unauthorized public/vehicle entry into work zone	Regulatory penalties / fines				authority approvals for the applicable traffic stage. Set-up and removal of traffic control arrangements shall be carried out only by trained and competent Traffic Management personnel under supervision of Traffic Management Engineer/Supervisor. All traffic diversions, lane closures, signs, cones, delineators, arrow boards, flashing beacons, barriers, and road markings shall be installed strictly in accordance with the approved drawings and RTA Work Zone Traffic Management requirements. Advance warning signs shall be positioned at required distances based on road classification, operating speed, and site conditions. Traffic setup activities shall be scheduled during approved working windows and coordinated with site operations to minimize exposure to live traffic and public inconvenience. Third-party certified Truck Mounted Attenuator (TMA) shall be deployed during installation, shifting, and dismantling of temporary traffic arrangements wherever required. Physical protection shall be provided between live traffic and workforce using approved barriers such as interlocked concrete barriers or water-filled barriers, properly aligned and linked to prevent displacement.	Conduct pre-opening inspection jointly by TTM Supervisor and HSE representative before releasing diversion to public traffic. Provide additional flashing beacons and illuminated arrow boards during night or poor visibility conditions. Carry out documented daily and post-impact inspection of barriers, signs, cones, and delineators. Suspend setup activity immediately in case of unsafe driver behavior, poor visibility, severe weather, or unapproved traffic stage change.				TTM Supervisor / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR DEWA- WD STORM WATER NETWORK

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SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							Adequate longitudinal and lateral safety buffer zones shall be maintained throughout the work area. Competent flagmen/banksmen shall be deployed at critical locations to guide movements during setup and to prevent unauthorized entry of vehicles into the work zone. Communication between traffic operatives, TMA drivers, and supervisors should be maintained through radios or other approved means. Night works or low-visibility operations shall be supported with adequate illumination, reflective traffic devices, high-intensity warning lamps, and reflective PPE to ensure visibility of both workers and control devices. All installed traffic control devices and barrier alignment shall be inspected immediately after setup, before opening any diversion to traffic, and thereafter at regular intervals to confirm integrity, visibility, and effectiveness. Any damaged, displaced, or missing device shall be rectified immediately. Safe pedestrian routes and crossing arrangements shall be maintained where public interface exists, and access to adjacent properties, emergency routes, and essential					

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) – Section C.8

RISK ASSESSMENT FOR DEWA- WD STORM WATERNETWORK

DATE: 23-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							services shall remain unobstructed at all times. Pre-task briefings shall be conducted before each setup activity to communicate traffic stage details, roles and responsibilities, live traffic hazards, emergency procedures, and stop-work criteria. Continuous supervision shall be ensured by Traffic Management Engineer, Site Engineer, and HSE personnel to monitor compliance and implement immediate corrective action where required. Mandatory PPE including high-visibility vest, helmet, safety shoes, and gloves shall be worn by all personnel involved in traffic management setup.					
3.	Survey & Setting Out	Slips, trips and falls; lone working; struck by plant/vehicles; exposure to live traffic; unsafe road crossing; unstable total station setup; protruding survey pins; hammering injuries; poor visibility	Personnel injury, fatality, traffic accident, property damage, inaccurate setting out leading to rework.	4 Likely	5 Catastroph	20 VERY HIGH	Survey activities shall be planned and coordinated with the construction and traffic management teams prior to commencement to ensure safe access and controlled working conditions. No survey work shall be carried out in active plant zones or near live traffic without proper coordination and approval. Lone work shall not be permitted, and minimum two trained personnel shall be deployed at all times to ensure communication and emergency support. All survey work near live traffic shall be carried out in accordance	Survey activities shall be carried out only after verification of approved drawings, latest IFC revisions, and coordination with utility drawings to avoid incorrect setting out and rework. Use of calibrated survey equipment shall be ensured, and calibration certificates shall be maintained and verified prior to deployment. Where survey is carried out near live traffic or high-risk zones, temporary physical protection (barriers or TMA where required) shall be provided to enhance	1 Remote	4 Likely	4 LOW	Chief Surveyor / Surveyor / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR DEWA- WD STORM WATERNETWORK										DATE: 23-03-2026		
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							with the approved Traffic Management Plan (TMP) and valid ROW permit. Temporary traffic control measures including cones, barriers, signage, and flagmen shall be implemented to segregate survey personnel from moving vehicles. Safe pedestrian access routes and controlled crossing points shall be established. Survey equipment including total station, tripod, prisms, and accessories shall be inspected prior to use to ensure proper condition and stability. Equipment shall be positioned on firm and level ground away from excavation edges, vehicle routes, and unstable surfaces. Tripods shall be securely fixed to prevent displacement due to vibration or wind. Protruding survey pins shall be immediately protected using caps or removed after use to eliminate puncture and trip hazards. Hammering of pins shall be carried out using appropriate tools with safe technique and sufficient clearance from other personnel. Work areas shall be maintained free from obstructions, and good housekeeping shall be ensured to prevent slips, trips, and falls.	protection of survey personnel. Permit-to-Work (PTW) shall be implemented where survey activities interface with high-risk areas such as live traffic, excavation zones, or lifting operations. Survey teams should maintain safe clearance from operating plants and equipment, and plant operators should be informed prior to commencement of survey activities within their working area. High-risk areas shall be identified and demarcated prior to survey work, and access shall be restricted to authorized personnel only. Use of GPS / total station with data verification shall be implemented to minimize manual error and improve accuracy of setting out. All survey points and benchmarks shall be protected, recorded, and verified periodically to avoid disturbance or loss due to ongoing construction activities. In case of poor visibility conditions (fog, dust, night), survey work shall be restricted or enhanced with				

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSG) - Section C.8

RISK ASSESSMENT FOR DEWA- WD STORM WATER NETWORK

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SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							Adequate lighting shall be provided during early morning, evening, or low-visibility conditions to ensure clear visibility of personnel and equipment. Pre-task toolbox talks shall be conducted to brief survey teams on hazards related to traffic, plant movement, and safe handling of equipment. Communication shall be maintained between survey teams, plant operators, and supervisors to avoid conflicts during operations. Mandatory PPE including safety helmet, high-visibility vest, and safety shoes shall be worn at all times. Additional PPE such as gloves and eye protection shall be used where required. Continuous supervision shall be ensured by Site Engineer and HSE personnel to monitor compliance and implement immediate corrective actions where necessary.	additional lighting and reflective measures. Emergency response arrangements should be in place, especially when working near live traffic or remote areas, including communication devices and quick access to first aid. Regular inspection and audit of survey activities shall be carried out by Site Engineer and HSE team to ensure compliance with approved procedures.				
4.	Trench Works (Identification & Protection of Existing Services & Utilities)	Striking underground services (DEWA, telecom, drainage), trench collapse, falling spoil, fall into excavation, exposure to live utilities (electric, gas), inaccurate service detection, plant interaction	Fatality, electrocution, explosion, serious injury, burial, service disruption, major fines, environmental contamination	4 Likely	5 Catastrophic	20 Very High	Trench works shall not commence without obtaining all required valid NOCs and approvals from relevant authorities (DEWA, DU/Etisalat, RTA, Drainage, Storm Water Works, etc.). Approved Permit to Work (PTW) and Permit-to-Dig shall be strictly implemented prior to any excavation activity. All existing utilities shall be	Utility detection shall be enhanced using Ground Penetrating Radar (GPR) or equivalent advanced detection methods where complexity or congestion of services exist, in addition to CAT & Genny verification. All utility identification and mapping shall be carried	1 Remote	5 Catastrophic	5 Medium	Construction Manager / Site Engineer / Supervisor / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

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				L	S	IRR			L	S	RRR	
							identified through approved utility drawings, records, and verification using CAT & Genny scanning by competent operators. Trial pits shall be carried out manually in controlled manner in sensitive zones to confirm exact depth, alignment, and type of services. Mechanical excavation shall be strictly restricted near identified services unless confirmed safe clearance is established. All utilities shall be clearly marked on ground using paint, flags, stakes, and warning tapes, and physical protection where exposed using suitable methods such as concrete slabs, sleeves, or protective covering. Dedicated utility coordinator or engineer shall verify and approve the location before proceeding with deeper excavation. Excavation areas shall be barricaded and provided with adequate warning signage to prevent unauthorized access. Safe access and egress shall be provided using ladders and fall protection measures shall be ensured around open trenches. Spoil shall be placed at a safe distance from trench edges to avoid collapse or falling material hazards.	out in line with PAS 128 standard (or equivalent best practice) for detection accuracy and classification of underground services. Hydro excavation (non-destructive digging) shall be considered in high-risk zones or where critical utilities (e.g., high-voltage DEWA, gas lines, fiber optics) are present to eliminate risk of mechanical damage. A formal Permit-to-Dig (PTD) system with multi-level authorization (Construction, Utility Coordinator, HSE) shall be implemented prior to commencement of each trial trench. Utility clearance certificate / verification checklist shall be completed and signed before allowing deeper excavation or mechanical work. Trial trench locations shall be physically verified and pegged by survey team prior to excavation to avoid deviation from approved alignment.				

HAZARD IDENTIFICATION & RISK ASSESSMENT

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				L	S	IRR			L	S	RRR	
							Continuous monitoring of trench stability shall be carried out by a competent person, and appropriate support systems such as shoring or trench boxes shall be implemented where required. No personnel shall enter trench deeper than permissible limits without proper support and inspection. Emergency response procedures shall be established and communicated, including immediate isolation of utilities in case of accidental strike. Standby arrangements for fire response, rescue, and first aid shall be maintained at site. Pre-task toolbox talks shall be conducted to brief workers on utility hazards, safe excavation practices, and emergency actions. Only trained and authorized personnel shall be involved in trial trench activities. Mandatory PPE including helmet, high-visibility vest, safety shoes, gloves, and additional protective equipment shall be used at all times. Continuous supervision shall be maintained by Site Engineer, Utility Coordinator, and HSE personnel to ensure strict compliance with safety requirements and immediate corrective action in case of unsafe conditions.	Exclusion zones shall be enhanced with rigid barriers (not only tapes) in high-risk or public interface areas to prevent accidental entry or collapse-related incidents. Dedicated standby utility drawings (latest approved + marked-up site drawings) shall be available at excavation point for reference during work execution. Close supervision by Utility Coordinator shall be mandatory during excavation near identified services, and no activity shall proceed without his clearance. Plant movement near trial trench shall be restricted, and minimum safe distance from excavation edges shall be enforced to avoid surcharge loading and collapse. Daily excavation inspection checklist shall be implemented and recorded by competent person, covering trench stability, utility exposure condition, access, and protection measures.				

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				L	S	IRR			L	S	RRR	
								Real-time communication protocol shall be established between excavation teams, plant operators, and supervision to immediately stop work in case of suspected utility encounter. Stop-work criteria shall be clearly defined, including: -Unidentified service encountered -Unexpected depth/location deviation -Signs of trench instability Utility damage or leakage Emergency utility isolation procedures shall be displayed and communicated, including immediate contact with relevant authorities (DEWA, telecom, etc.) in case of strike. Environmental protection measures shall be implemented to control contamination in case of utility damage (e.g., water, sewage, fuel lines).				
5.	Arrangement for Access for Storm Water Works Pipeline	Vehicle-pedestrian conflict with live traffic; unauthorized entry into work zone; obstruction of emergency access; unsafe entry/exit points; damage to	Fatality, serious injury, traffic accidents, delay in emergency response, utility damage, regulatory penalties, public complaints	4 Likely	5 Catastrophic	20 Very High	Access arrangements shall be established only after obtaining approved Traffic Management Plan (TMP), valid RTA ROW permit, and all required authority NOCs. No access point shall be created or used without formal approval and site verification. Access routes shall	Trial setup / mock-up of access point shall be carried out (where required) and verified by Traffic Management Engineer prior to full implementation to ensure compliance with approved TMP.	1 Remote	5 Catastrophic	05 Medium	TTM Manager / Site Engineer / Supervisor

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				L	S	IRR			L	S	RRR	
		existing utilities during access formation; poor visibility at access points					be planned in coordination with Traffic Management Team to minimize interaction with live traffic and public movement. Dedicated entry and exit points shall be clearly defined, properly marked, and controlled at all times. Temporary traffic control measures including advance warning signs, directional signage, cones, barriers, and flashing beacons shall be installed in accordance with RTA Work Zone Traffic Management Manual requirements. Adequate buffer zones shall be maintained based on road speed and site conditions. Competent flagmen/banksmen shall be deployed at all access points to control movement of construction vehicles and ensure safe interaction with traffic and pedestrians. Entry and exit of vehicles shall be coordinated to avoid congestion and sudden movements. Drivers shall be briefed on approved access routes and site-specific traffic rules prior to entry. Work areas shall be physically segregated from live traffic using suitable barriers such as concrete or water-filled barriers to prevent unauthorized access and protect workforce. Unauthorized personnel and vehicles shall be strictly prohibited from entering the work	Deployment of Truck Mounted Attenuator (TMA) shall be considered at high-speed roads or critical access points during entry/exit operations and during installation/removal of access arrangements. Dedicated access marshal / gate controller shall be assigned at each access point to control and log vehicle movement, ensuring controlled entry and exit at all times. Access log registers shall be maintained to record movement of plant, equipment, and delivery vehicles for better control and traceability. Speed control measures such as rumble strips, speed limit signage, or temporary speed calming devices shall be implemented at entry/exit points to reduce vehicle speed. Additional lighting arrangements (tower lights or fixed lighting) should be provided at access points during night operations to enhance visibility of				

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				L	S	IRR			L	S	RRR	
							zone, and access control measures shall be enforced. Emergency access routes should be clearly identified, maintained unobstructed at all times, and regularly inspected to ensure immediate availability for emergency vehicles. No material storage, plant parking, or obstruction shall be allowed within emergency access routes. Access routes shall be inspected for underground utilities prior to use, and protective measures shall be implemented to avoid damage to existing services. Ground conditions shall be verified to ensure safe movement of vehicles and equipment. Pre-task briefings shall be conducted to communicate access arrangements, hazards, and control measures to all personnel. Continuous monitoring and daily inspection shall be carried out by Traffic Management Team, Site Engineer, and HSE personnel to ensure compliance and immediate rectification of unsafe conditions. Mandatory PPE including high-visibility vest, helmet, and safety shoes shall be worn by all personnel working near access and traffic interface areas. Additional lighting and reflective devices shall	vehicles, pedestrians, and control devices. Reflective road studs / markers shall be installed at critical access zones to enhance lane visibility, especially during night or adverse weather conditions. Access route condition monitoring checklist shall be implemented daily to verify ground stability, signage condition, barrier alignment, and visibility. Temporary access routes crossing utilities shall be provided with load distribution measures such as steel plates or protective slabs to prevent damage to underlying services. Clear demarcation of pedestrian exclusion zones shall be ensured near access points, with additional signage and physical barriers to prevent pedestrian-vehicle interaction. Coordination with nearby contractors and stakeholders shall be ensured to avoid conflicting access				

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				L	S	IRR			L	S	RRR	
							be provided during night or low-visibility conditions.	arrangements or overlapping routes. Emergency vehicle priority protocols shall be established, ensuring immediate clearance of access routes in case of emergency. Stop-work criteria shall be defined, including: -Poor visibility (fog, dust storm, night without lighting) -Damaged or missing traffic control device. -Unsafe traffic behavior or congestion -Unstable access route condition Regular joint inspection with Traffic Management Team and HSE shall be conducted to verify compliance with approved TMP and RTA standards.				
6.	Dewatering Work (Commissioning, Operation & Monitoring)	Pump failure, electrical short circuit, fire hazard, pipeline rupture, uncontrolled discharge, flooding, oil/fuel spill, excessive noise, unstable excavation due to water accumulation	Minor to major injury, flooding of excavation, environmental contamination, property damage, public nuisance, regulatory fines, work delay	4 Likely	3 Moderate	12 HIGH	Dewatering activities shall not commence without approved dewatering layout, discharge plan, and relevant authority approvals in line with project specifications and CEMP requirements. All discharge points shall be verified and approved prior to operation to ensure no adverse environmental impact.	Automatic level sensors / float switches shall be installed to activate pumps automatically in case of rising water levels, ensuring continuous control without manual delay. Backup power supply (generator with auto-start system) shall be provided to ensure uninterrupted	1 Remote	3 Moderate	3 LOW	Pump Operator / Supervisor / Electrician / Electrical Engineer / Construction Manager

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				L	S	IRR			L	S	RRR	
							Dewatering pumps, generators, pipelines, and associated equipment shall be inspected, tested, and certified prior to commissioning. Only competent and trained personnel shall operate and monitor the system. Backup pumps and standby generators shall be maintained at site to ensure continuity in case of equipment failure and to prevent flooding of excavation. Electrical connections shall be carried out by certified electricians using properly protected distribution boards with ELCB and earthing systems. All cables shall be routed safely to avoid water contact and mechanical damage. Regular inspection of electrical systems should be conducted to prevent short circuits and fire hazards. Fuel storage for generators shall be maintained in designated bunded areas with proper containment and spill kits readily available. Refueling activities shall be controlled and supervised to avoid spillage and fire risk. Discharge pipelines shall be properly secured and inspected to prevent leakage, rupture, or uncontrolled discharge. Measures such as sediment control, filter media, or rock fill shall be provided at discharge points to prevent	dewatering in case of power failure. A standby pump (minimum 100% backup capacity) shall be maintained in ready condition with pre-connected hoses for immediate deployment. Discharge flow monitoring shall be carried out to ensure system efficiency and detect abnormal discharge conditions or blockages. Regular flushing and cleaning of discharge pipelines shall be implemented to prevent clogging, sediment buildup, and pressure increase. Discharge water shall be tested or visually inspected for contamination (oil, silt, debris) prior to release, ensuring compliance with environmental requirements. Silt traps / sedimentation tanks shall be installed where required to prevent discharge of suspended solids into drainage systems. Dewatering log sheet / monitoring record shall be				

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				L	S	IRR			L	S	RRR	
							erosion and environmental contamination. Excavation areas shall be continuously monitored for water accumulation, soil stability, and potential collapse risks. Immediate action shall be taken in case of abnormal water levels or system malfunction. Noise levels from pumps and generators shall be controlled using maintained equipment and, where necessary, noise barriers to minimize disturbance to workers and surrounding environment. Pre-task briefings shall be conducted to communicate hazards, emergency procedures, and operational controls. Emergency response arrangements shall be established for pump failure, flooding, electrical hazards, and spill incidents. Mandatory PPE including helmet, safety shoes, gloves, high-visibility vest, and hearing protection (where required) shall be used. Continuous supervision shall be ensured by Supervisor, Electrical Engineer, and HSE personnel to maintain safe and compliant operations.	maintained to record pump operation time, water levels, inspection status, and any abnormal observations. Daily inspection checklists should be implemented covering pumps, electrical systems, hoses, bunding, and discharge points. Access to dewatering equipment shall be restricted and properly barricaded to prevent unauthorized interference or accidental damage. Noise monitoring shall be conducted where pumps/generators operate near public or sensitive areas, and corrective measures shall be implemented if limits are exceeded. Vibration monitoring should be considered where dewatering operations are close to sensitive structures or utilities. Emergency overflow route or contingency discharge arrangement shall be identified to manage excess water in case of system overload.				

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				L	S	IRR			L	S	RRR	
								Stop-work / shutdown criteria shall be clearly defined, including: -Pump failure without backup availability -Electrical fault or unsafe condition -Uncontrolled discharge or leakage -Excessive water accumulation affecting trench stability Coordination with excavation and construction teams shall be ensured to align dewatering operations with ongoing activities and avoid conflict or unsafe conditions.				
7.	Excavation & Formation Preparation	Trench collapse / sidewall instability; fall into excavation; struck by moving plant; water ingress / flooding; underground utility strike; confined space conditions; unstable soil; falling materials; dust exposure	Fatality (burial/drowning), serious injury, electrocution, service damage, environmental contamination, regulatory penalties, project delay	4 Likely	5 Catastrophic	20 Very High	Excavation works shall not commence without approved Method Statement, valid Permit-to-Dig, and confirmed utility clearance from relevant authorities. All underground services shall be identified, marked, and verified prior to excavation through approved drawings, trial pits, and detection methods. Excavation shall be carried out in controlled stages and limited open trench lengths to minimize exposure. Trench sides shall be sloped, benched, or supported using appropriate shoring systems or trench boxes based on soil condition and excavation depth as per geotechnical recommendations.	Trial trench / potholing shall be completed and verified prior to bulk excavation to confirm exact location, depth, and alignment of all underground services. Excavation works shall be supported by approved excavation permit system (Permit-to-Dig) with daily revalidation before continuation of work. Excavation risk assessment shall be reviewed daily based on changing site conditions such as soil type, weather, groundwater level, and nearby activities.	1 Remote	5 Catastrophic	05 MEDIUM	Construction Manager / Site Engineer / Foreman / HSEO

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				L	S	IRR			L	S	RRR	
							No personnel shall be permitted to enter excavation deeper than allowable limits without proper inspection, support, and authorization by a competent person. Daily inspection of excavation, soil stability, groundwater condition, and protective systems shall be carried out and recorded prior to allowing entry. Excavation areas shall be barricaded with warning signs, edge protection, and controlled access. Safe access and egress shall be provided using ladders placed at suitable intervals. Spoil materials shall be placed at a safe distance from trench edges to prevent collapse or falling hazards. Dewatering arrangements shall be implemented to control groundwater and prevent flooding. Continuous monitoring shall be conducted to ensure excavation remains dry and stable. Emergency response measures shall be in place for flooding or collapse scenarios. Plant and equipment movement shall be controlled with designated access routes, and a competent banksman shall guide all operations near excavation areas. A safe distance shall be maintained between heavy equipment and trench edges to avoid surcharge loading and collapse.	Protective systems design (shoring/trench box) should be verified and, where required, supported by engineering calculations or geotechnical recommendations for deeper excavations. Load restriction zones shall be established along trench edges to prevent surcharge loading from plant, materials, or traffic. Physical edge protection (guardrails or rigid barriers) shall be installed for deep excavations, especially in areas with worker or public interface. Gas detection (multi-gas detectors) shall be used before and during entry into deep or confined excavations to monitor oxygen level and presence of hazardous gases. Emergency rescue equipment (tripod, harness, rescue kit) shall be available for deep excavation or confined space conditions. Weather monitoring systems shall be implemented, and				

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				L	S	IRR			L	S	RRR	
							Confined space conditions shall be assessed where applicable, and additional controls shall be implemented including gas monitoring, ventilation, and rescue arrangements before entry into deep or restricted excavations. Dust generation shall be controlled through water spraying and proper housekeeping. Work areas shall be maintained clean and organized to prevent slips, trips, and falls. Pre-task toolbox talks shall be conducted to communicate hazards, control measures, and emergency procedures. Only trained and competent personnel shall be involved in excavation work. Mandatory PPE including helmet, high-visibility vest, safety shoes, gloves, and respiratory protection (if required) shall be used. Continuous supervision shall be ensured by Site Engineer, Foreman, and HSE personnel to maintain compliance and immediately address unsafe conditions.	excavation activities shall be suspended during heavy rain, sandstorms, or adverse conditions affecting stability. Daily excavation inspection checklists shall be implemented and recorded by a competent person before start of work and after any significant change (rainfall, vibration, etc.). Survey control points and excavation alignment shall be verified periodically to avoid deviation and ensure accuracy of formation levels. Barricading shall be enhanced using rigid barriers (not only warning tape) in high-risk or public areas to prevent accidental entry or falls. Safe access points (ladders/ramps) shall be installed at regular intervals (max. 7-10m spacing) and secured properly. Spoil management plan shall be implemented to control location, height, and distance of excavated material.				

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				L	S	IRR			L	S	RRR	
								Stop-work criteria shall be clearly defined, including: -Signs of trench instability (cracks, bulging, sloughing) -Water accumulation or dewatering failure -Discovery of unidentified utility -Failure or damage of shoring system -Unsafe plant movement near trench Coordination with adjacent activities (lifting, traffic, dewatering) shall be ensured to avoid conflicting operations. Regular HSE inspection and audit shall be conducted to verify compliance with excavation safety requirements and MSTG standards.				
8.	Laying of GRP Pipes (Storage, Transportation, Geotextile & Granular Bedding, Installation)	Pipe drop or rolling; lifting failure; struck-by during unloading; manual handling strain; unstable trench base; improper bedding; misalignment; working near excavation edges; vehicle movement during delivery	Crush injury, amputation, musculoskeletal injury, pipe damage, installation defects, rework, property damage	4 Likely	5 Catastrophic	20 Very High	GRP pipes shall be delivered, stored, handled, and installed strictly in accordance with manufacturer recommendations, approved Method Statement, and project specifications. All pipes shall be inspected upon delivery to ensure they are free from defects, damage, or deformation prior to use. Pipes shall be stored in designated, level, and stable areas away from traffic routes. They shall be supported on timber sleepers with	Lifting plan specific to GRP pipe handling shall be prepared and approved, considering pipe weight, length, lifting points, and sling arrangement to avoid damage and imbalance. Only manufacturer-approved lifting methods (e.g., wide webbing slings, no chain contact) should be used to prevent surface damage or deformation of GRP pipes.	1 Remote	4 Major	4 LOW	Lifting Supervisor / Site Engineer / Foreman / HSEO

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				L	S	IRR			L	S	RRR	
							proper spacing and secured using wedges or chocks to prevent rolling. Stacking height shall be controlled as per manufacturer guidelines to maintain stability. Transportation of pipes shall be carried out using suitable flatbed trailers, and loading/unloading shall be performed using certified lifting equipment such as cranes or HIAB. Only certified lifting accessories (webbing slings, shackles) shall be used, and lifting shall be supervised by a competent lifting supervisor with proper communication between operator and banksman. Manual handling of pipes shall be minimized, and mechanical means shall be used wherever practicable. Where manual handling is required, team lifting and proper techniques should be applied to avoid strain injuries. Excavation trenches shall be inspected and approved prior to pipe installation to ensure stability, correct formation level, and absence of water accumulation. Granular bedding shall be prepared to specified thickness and compacted to required density before pipe laying. Exclusion zones shall be established around lifting and pipe installation areas to prevent unauthorized	Taglines shall be used during lifting to control pipe movement and prevent uncontrolled rotation or swing. Trial lift shall be conducted for large diameter or heavy pipes to verify lifting arrangement and stability before full operation. Pipe handling inspection checklist shall be implemented to verify conditions before installation (no cracks, deformation, or coating damage). Alignment and level shall be verified using survey instruments before final placement to avoid rehandling or correction under unsafe conditions. Joining inspection and approval shall be carried out by QA/QC Engineer before proceeding to backfilling to ensure proper sealing and alignment. Bedding compaction and level verification records (FDT/inspection) shall be maintained prior to pipe placement to ensure stability and load distribution.				

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				L	S	IRR			L	S	RRR	
							access. No personnel shall be allowed beneath suspended loads, and safe working distances shall be maintained at all times. Pipes shall be aligned accurately using survey control, and proper jointing procedures shall be followed as per manufacturer guidelines. Pipe ends shall be protected, cleaned, and lubricated before connection to ensure proper sealing and prevent leakage. Working near trench edges shall be controlled with edge protection and fall prevention measures shall be implemented. Access into trenches shall be provided through safe ladders and controlled entry points. Pre-task briefings shall be conducted to communicate hazards, lifting procedures, and installation requirements. Continuous supervision shall be maintained by Site Engineer, Foreman, and HSE personnel. Mandatory PPE including helmet, safety shoes, gloves, and high-visibility vest shall be worn at all times. Additional PPE such as fall protection equipment shall be used where required.	Load restrictions shall be enforced over installed pipes, and no heavy equipment shall pass over pipeline until adequate cover is achieved. Temporary support or side packing shall be provided to stabilize pipes during installation and prevent rolling or displacement. Dedicated access and movement plans for delivery vehicles shall be implemented to avoid congestion and unsafe interaction with site activities. Communication protocols (radio/hand signals) shall be established between lifting team, banksman, and supervisor for coordinated operation. Weather conditions (wind) shall be monitored during lifting operations, and lifting shall be suspended if conditions become unsafe. Stop-work criteria shall be defined, including: -Unstable trench condition -Improper bedding or water accumulation -Damaged pipe detected				

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				L	S	IRR			L	S	RRR	
								-Unsafe lifting conditions or equipment failure -Poor visibility or adverse weather				
9.	Installation of Precast Slab / Protection Slab	Lifting failure; load drop; crushing during placement; slab misalignment; struck-by suspended load; fall into trench; working near excavation edges; unstable ground for crane setup	Fatality, crush injury, amputation, major property damage, structural defect, rework	4 Likely	5 Catastrophic	20 Very High	Installation of precast slabs shall be carried out strictly in accordance with approved lifting plan, Method Statement, and shop drawings. No lifting activity shall be performed without valid Permit-to-Work and approved lifting arrangement. Only certified cranes/HIAB and lifting accessories (slings, shackles, hooks) with valid third-party certification shall be used. All lifting equipment shall be inspected prior to use to ensure integrity and suitability for the load. Load weight shall be verified, and lifting capacity shall not be exceeded. Ground conditions at crane setup location shall be assessed and prepared to ensure stability. Outriggers shall be fully extended on firm ground with suitable support (steel plates or timber mats) to prevent overturning. A full exclusion zone shall be established around the lifting and installation area, and no personnel shall be allowed under or near suspended loads. Competent lifting supervisors, operators, and banksmen shall be present at all times to control and coordinate lifting activities.	Detailed lifting plan (including load chart, radius, ground bearing pressure calculation) shall be reviewed and approved prior to commencement of lifting activities. Trial lift shall be conducted to verify load balance, lifting points, rigging arrangement, and crane stability before actual placement. Lifting accessories selection shall be based on load weight and configuration, ensuring proper sling angle and capacity to avoid overstraining. Tagline control shall be mandatory for all precast slab lifting to prevent uncontrolled rotation and ensure precise positioning. Pre-lift checklist shall be implemented covering crane condition, rigging, ground stability, exclusion zone, weather, and communication readiness. Ground bearing pressure verification shall be carried	1 Remote	5 Catastrophic	05 Medium	Site Engineer / Supervisor / Lifting Supervisor / Surveyor

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				L	S	IRR			L	S	RRR	
							Slabs shall be lifted only from designated lifting points using approved methods to prevent imbalance or sudden movement. Taglines shall be used where necessary to control load rotation and ensure safe positioning. Excavation/trench shall be inspected and verified stable prior to slab installation. Edge protection and safe access arrangements should be provided to prevent falls. No personnel shall enter trench during lifting or placement of slabs. Surveyors shall verify alignment, level, and positioning before final placement to avoid rehandling or adjustment under unsafe conditions. Pre-task briefings shall be conducted to communicate lifting procedures, hazards, and emergency actions. Weather conditions including wind speed shall be monitored, and lifting shall be suspended during unsafe conditions. Mandatory PPE including helmet, gloves, safety shoes, and high-visibility vests shall be worn. Additional fall protection should be used where required. Continuous supervision shall be maintained by Lifting Supervisor, Site Engineer, and HSE personnel.	out to ensure crane setup area can safely support imposed loads, especially near excavations. Load paths shall be planned and kept clear, ensuring no obstruction or personnel exposure during movement of slab from pickup to installation point. Temporary support or guide system should be provided where required to assist in safe placement and alignment of slabs. Survey verification (line, level, and orientation) shall be carried out before and after placement to ensure correct positioning and avoid rehandling. Joint inspection and approval shall be completed before allowing backfilling or further activities over installed slabs. No simultaneous conflicting activities (excavation, lifting, traffic movement) shall be allowed within the lifting zone. Dedicated communication systems (radio or standard hand signals) shall be				

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				L	S	IRR			L	S	RRR	
								maintained between crane operator, banksman, and lifting supervisor. Wind speed monitoring devices (anemometer) shall be used where required, and lifting shall be stopped when limits exceed manufacturer recommendations. Enhanced barricading using rigid barriers shall be provided in high-risk or public interface areas (tape). Stop-work criteria shall be clearly defined, including: -Crane instability or ground settlement -Improper rigging or lifting imbalance -Excessive wind or poor visibility -Uncontrolled load movement -Unsafe trench condition Regular lifting inspection shall be conducted by HSE and lifting supervisor to verify compliance with lifting plan and standards.				
10.	Concrete Surround, Thrust Blocks & Protection Slab	Formwork collapse; struck-by concrete during pouring; contact with wet concrete (cement	Major injury, chemical burns, crush injury, structural defects, rework, property damage	3 Rare	4 Major	12 High	Concrete works shall be carried out strictly in accordance with approved Method Statement, shop drawings, and project specifications to ensure both safety and structural integrity.	Formwork design calculations and shop drawings shall be reviewed and approved prior to installation, especially for	1 Remote	4 Major	4 Low	Foreman. Chargehand and engineer.

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				L	S	IRR			L	S	RRR	
		burns); manual handling of rebar/formwork; improper placement; vibration equipment hazards; working near excavation edges					Only approved ready-mix concrete shall be used, and on-site uncontrolled mixing shall not be permitted. Formwork systems should be designed, installed, and inspected prior to pouring to ensure adequate strength, alignment, and stability. No concrete pouring shall commence until formwork inspection and approval are completed by the responsible engineer and consultant. Concrete pouring shall be carried out in a controlled and sequential manner to avoid overloading or displacement of formwork and adjacent structures. Workers shall be positioned safely to avoid exposure to moving equipment and falling materials. Handling of reinforcement, shuttering materials, and accessories shall be properly planned to minimize manual handling risks. Mechanical aids shall be used wherever practicable, and team lifting techniques shall be applied where necessary. Vibrators and associated electrical equipment shall be inspected prior to use and operated only by trained personnel. Electrical connections shall be safe, properly insulated, and protected from water ingress.	deep excavation or large thrust blocks. Pre-pour inspection checklist (ITP-based) shall be completed and signed by Site Engineer and QA/QC Engineer before concrete pouring. Concrete pour permit system shall be implemented to ensure all safety, quality, and formwork conditions are verified prior to commencement. Trial mix / slump test and concrete quality verification shall be carried out at site to ensure compliance with project specifications. Continuous monitoring during pouring shall be maintained to detect any formwork movement, leakage, or instability, and immediate action shall be taken if abnormality is observed. Pouring rate and sequence shall be controlled to prevent excessive pressure on formwork and avoid structural failure.				

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				L	S	IRR			L	S	RRR	
							Work areas shall be barricaded and access restricted during pouring operations to prevent unauthorized entry. Adequate working platforms should be provided where required, especially near excavation edges, to prevent falls. Contact with wet concrete shall be minimized, and immediate washing facilities shall be available to prevent skin irritation or chemical burns. Proper curing procedures shall be followed after placement to maintain quality and durability. Pre-task briefings shall be conducted to communicate hazards, sequence of work, and control measures. Continuous supervision shall be ensured by Site Engineer and Foreman during concrete activities. Mandatory PPE including gloves, safety goggles, helmet, safety shoes, and protective clothing shall be worn at all times. Additional PPE such as rubber boots and chemical-resistant gloves should be used when handling wet concrete.	Dedicated safe access platforms / working platforms shall be provided with guardrails where personnel are required to work at height or near excavation edges. Concrete pump line and hose routing shall be secured and positioned to avoid whipping, sudden movement, or obstruction to workers. Rebar protection (capping of exposed ends) shall be provided to prevent impalement injuries. Isolation of pouring zones shall be ensured with rigid barricading in high-risk or public interface areas. Vibrator usage duration shall be controlled to avoid over-vibration, segregation of concrete, and structural defects. Post-pour curing and protection measures shall be implemented immediately to prevent cracking and ensure structural integrity. Inspection of completed works before backfilling				

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				L	S	IRR			L	S	RRR	
								shall be carried out and approved by QA/QC and consultant. Stop-work criteria shall be clearly defined, including: -Formwork instability or movement -Excessive leakage of concrete -Unsafe working platform or access -Equipment failure (vibrator/pump) -Adverse weather affecting pour quality Coordination with adjacent activities (excavation, lifting, traffic) shall be ensured to avoid simultaneous conflicting operations. Regular HSE and QA/QC inspection shall be conducted to verify compliance with method statement and safety requirements.				
11.	Hot Work (Grinding / Welding / Cutting)	Fire outbreak due to sparks; explosion near flammable materials; burns from hot metal; welding fumes exposure; arc flash (eye injury); gas cylinder leakage; electrical shock; poor ventilation; ignition of nearby materials	Fatality, severe burns, explosion, fire damage, respiratory illness, property damage, work suspension	4 Likely	5 Catastrophic	20 Very High	Hot work activities shall not be carried out without a valid Hot Work Permit issued for the specific location, duration, and task. Work shall be strictly controlled under Permit-to-Work system with full compliance with site procedures and supervision. Prior to commencement, the work area shall be thoroughly inspected	Hot Work Risk Assessment shall be reviewed and attached to the permit for each activity, considering surrounding hazards and location-specific risks. Gas testing (O ₂ , LEL, toxic gases) shall be carried out prior to and during hot work in enclosed or confined	1 Remote	5 Catastrophic	05 Medium	Hot Work Supervisor / Permit Issuer / Fire Watcher / HSE

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				L	S	IRR			L	S	RRR	
							and cleared of all flammable and combustible materials within a minimum safe radius. Where removal is not possible, suitable fire-resistant coverings or blankets shall be used to protect adjacent materials and structures. Only trained and certified personnel shall be allowed to perform welding, cutting, or grinding operations. All equipment including welding machines, cables, electrodes, and cutting tools shall be inspected prior to use to ensure proper condition and safe operation. Gas cylinders shall be stored upright, secured with chains, and fitted with regulators and flashback arrests. Cylinders shall be kept away from heat sources and checked for leakage prior to use. A dedicated fire watcher shall be assigned during hot work and shall remain in place for a minimum of 30 minutes after completion to monitor for any potential fire hazards. Suitable fire extinguishers (CO₂, DCP) and fire blankets shall be readily available at work location. Work areas shall be properly ventilated, especially in enclosed or confined spaces, to prevent accumulation of toxic fumes. Where required, mechanical ventilation or	areas to ensure safe atmospheric conditions. Continuous fire watch monitoring system should be implemented, including documented fire watch checklist and periodic inspection during and after work. Extended fire watch duration (up to 60 minutes) shall be applied for high-risk areas or where combustible materials are present. Hot work areas shall be clearly demarcated with rigid barriers (not only tape) in high-risk or public interface locations. Dedicated spark containment measures such as spark arrestors, welding screens, or protective enclosures shall be used to prevent spread of sparks and slag. Isolation of nearby utilities and flammable systems shall be ensured prior to hot work (e.g., fuel lines, gas lines, electrical sources). Portable gas cylinders shall be transported using trolleys				

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				L	S	IRR			L	S	RRR	
							extraction systems shall be provided. Electrical connections shall be properly insulated and earthed, and equipment shall be protected from moisture and damage. Work area shall be barricaded and access restricted to authorized personnel only. Sparks and slag shall be controlled to prevent spread to adjacent areas. Pre-task toolbox talks shall be conducted to communicate fire hazards, emergency procedures, and safe work practices. Emergency response arrangements including fire alarm, evacuation routes, and first aid shall be established. Mandatory PPE including welding helmet/face shield, flame-resistant gloves, protective clothing, safety shoes, and respiratory protection (where required) shall be used. Continuous supervision shall be maintained by Supervisor and HSE personnel to ensure strict compliance and immediate corrective action in case of unsafe conditions.	and secured at all times, and separation distance between oxygen and fuel gas cylinders shall be maintained as per standards. Flashback arrests and non-return valves shall be checked daily and maintained in good condition. Hot work in confined spaces shall be strictly controlled under confined space permits, including ventilation, standby rescue team, and continuous monitoring. Electrical welding machines shall be connected through proper distribution boards with ELCB protection, and cable routing shall be elevated or protected to prevent damage. Work shall be immediately stopped in case of unsafe conditions, including: -Presence of flammable vapors -Failure of fire protection equipment -Gas leakage or abnormal smell -Inadequate ventilation				

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				L	S	IRR			L	S	RRR	
								-Unsafe weather (wind spreading sparks) Hot work permit validity shall be limited to shift duration only, and revalidation shall be required for continuation. Regular inspection and audit of hot work activities shall be conducted by HSE team to ensure compliance with permit conditions and safety standards. Emergency drills specific to fire scenarios shall be conducted periodically to ensure preparedness of workers and response team.				
12.	Waterproofing on Chambers (Internal & External)	Chemical exposure (bitumen, epoxy, membranes); toxic fumes in confined space; poor ventilation; slips on wet surfaces; fire hazard from flammable materials; working at height (chamber edges); skin/eye contact	Chemical burns, respiratory illness / asphyxiation, fire/explosion, slips/falls, environmental contamination	4 Likely	4 Major	16 Very High	Waterproofing activities shall only be carried out in accordance with approved Method Statement and Material Safety Data Sheets (MSDS) for all chemicals used. No work shall commence without proper material approval and hazard communication to all personnel involved. Work inside chambers shall be assessed for confined space conditions, and entry shall only be permitted under a valid confined space Permit-to-Work with all required safety arrangements in place. Adequate ventilation shall be provided using mechanical exhaust	Material approval and MSDS review checklist shall be implemented prior to use of any waterproofing chemical to verify hazards, compatibility, and safe handling requirements. Chemical risk assessment (COSHH-style) shall be prepared and communicated to workers, highlighting exposure limits, handling procedures, and emergency actions. Pre-entry confined space checklist shall be completed	1 Remote	4 Major	4 Low	Site Eng / Supervisor / HSE

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				L	S	IRR			L	S	RRR	
							fans or blowers to ensure safe oxygen levels and removal of harmful vapors. Continuous gas monitoring shall be carried out where applicable. Only trained and competent personnel shall carry out waterproofing work, and all workers shall be briefed on chemical hazards, safe handling procedures, and emergency response actions prior to commencement. Chemicals and waterproofing materials shall be stored in designated, well-ventilated areas away from ignition sources. Flammable materials shall be handled carefully, and appropriate fire prevention measures shall be implemented including availability of suitable fire extinguishers. Application areas shall be barricaded and access restricted to authorized personnel only. Work surfaces shall be kept clean and dry as far as practicable to prevent slip hazards, and immediate cleaning shall be carried out in case of spills. Proper application tools and techniques shall be used to minimize splashing, spillage, and excessive exposure. Handling of hot bitumen or liquid materials shall be	and signed prior to entry, including verification of ventilation, gas levels, rescue arrangements, and communication systems. Continuous gas monitoring devices (multi-gas detector) shall be used throughout the activity to monitor oxygen level and presence of toxic or flammable vapors. Standby rescue personnel shall be assigned at all times during confined space work with ready access to rescue equipment (tripod, harness, lifeline). Use of intrinsically safe equipment shall be ensured in confined spaces where flammable vapors may be present. Secondary containment measures (drip trays / bunding) shall be provided during application of liquid chemicals to prevent spillage and environmental contamination. Dedicated chemical storage and handling area shall be provided with proper				

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				L	S	IRR			L	S	RRR	
							controlled to prevent burns and fire risk. Pre-task toolbox talks shall be conducted to communicate hazards, PPE requirements, and emergency procedures. Emergency eyewash stations and first aid facilities shall be readily available at work locations. Mandatory PPE including chemical-resistant gloves, safety goggles/face shield, respirators (where required), protective clothing, helmet, and safety shoes shall be used at all times. Continuous supervision shall be ensured by Site Engineer, Supervisor, and HSE personnel to monitor compliance and take immediate corrective action in case of unsafe conditions.	labeling, segregation, and spill control arrangements. Hot work restriction shall be strictly enforced near waterproofing activities, especially when flammable materials (bitumen, solvents) are in use. Application method control shall be implemented to avoid excessive buildup, splashing, or uncontrolled spreading of chemicals. Slip-resistant measures (temporary mats, anti-slip coatings) shall be used in areas where wet surfaces are unavoidable. Ventilation effectiveness shall be periodically checked to ensure continuous removal of fumes, especially in deep or enclosed chambers. Exposure time control (work-rest cycle) shall be implemented to limit worker exposure to chemical fumes and heat. Stop-work criteria shall be clearly defined, including:				

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				L	S	IRR			L	S	RRR	
								-Unsafe gas levels (low oxygen / high LEL / toxic gases) -Failure of ventilation system -Excessive chemical vapor accumulation -Spill or uncontrolled release of chemicals -Fire risk or ignition source detected Environmental protection measures shall be implemented to prevent contamination of soil, drainage, or nearby water bodies. Regular inspection of waterproofing activities shall be carried out by HSE and QA/QC team to ensure compliance with MSDS and method statement.				
13.	Backfilling & Compaction	Trench collapse during backfilling; struck-by moving plant; compactor vibration hazards; damage to installed pipeline/services; dust generation; uneven compaction; working near excavation edges	Personal Injury, crush injury, pipeline damage, rework, service failure, respiratory issues, property damage.	4 Likely	4 Major	16 Very High	Backfilling and compaction activities shall be carried out strictly in accordance with the approved Method Statement, project specifications, and consultant requirements to ensure structural stability and safety of surrounding works. Backfilling shall commence only after inspection and formal approval of installed pipeline, bedding, and associated work by the Site Engineer and QA/QC team.	Backfilling approval checklist (ITP-based) shall be completed prior to commencement, ensuring pipeline installation, bedding, alignment, and jointing are fully approved. Compaction method statement compliance verification shall be conducted on site to ensure correct equipment selection,	1 Remote	4 moderates	4 Low	Site Engineer / QAQC Engineer / HSE Office

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				L	S	IRR			L	S	RRR	
							No backfilling shall proceed without confirmed readiness and clearance. Backfilling shall be conducted in controlled layers of specified thickness using approved materials, with each layer compacted to achieve the required density. Field Density Tests (FDT) shall be carried out and approved by the consultant prior to proceeding with subsequent layers to ensure compliance with project specifications. Only trained and competent operators shall be permitted to operate compaction equipment, and their competency shall be verified prior to deployment. All compactors shall be inspected daily before use, including checks on throttle control, vibration system, fuel lines, and overall mechanical condition to ensure safe operation. Compaction sequence shall be strictly followed to prevent uneven settlement or structural damage. Installed pipelines and utilities shall be adequately protected during backfilling, and no heavy equipment shall be allowed over exposed services until sufficient cover is achieved. Movement of plants and equipment shall be controlled through designated access routes, and trained banksmen shall be deployed	layer thickness, and compaction sequence. Field Density Test (FDT) frequency shall be defined and monitored as per specification, and records shall be maintained for consultant verification. Trial compaction (test strip) shall be carried out where required to confirm compaction method and achieve required density. Pipeline protection measures such as sand surround, warning tape, or protective layers shall be implemented prior to compaction. Load restriction shall be enforced to prevent movement of heavy equipment over pipeline until minimum cover depth is achieved. Temporary physical protection (plates or barriers) shall be provided over critical utilities or pipelines where equipment movement is unavoidable. Ground condition monitoring checklist shall be implemented daily to assess				

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				L	S	IRR			L	S	RRR	
							to guide operations near trenches and personnel. A minimum safe distance shall be maintained from excavation edges to prevent collapse or edge failure. Where work is required near trench edges, additional protection or temporary support shall be provided. Dust generation shall be controlled through regular water spraying, particularly during dry and windy conditions, in compliance with the approved Construction Environmental Management Plan (CEMP). Exposure to vibration (HAVS) shall be minimized through limitation of continuous operation time and job rotation. Operators shall be monitored for signs of fatigue and discomfort. Fuel used for compaction equipment shall be stored in approved containers within designated areas away from ignition sources. Any fuel leakage shall be immediately controlled using spill kits to prevent environmental contamination. Excavation edges and surrounding ground conditions shall be continuously monitored to ensure stability during backfilling operations. Personnel shall not be allowed in unsafe zones or under operating equipment.	stability, settlement, or any abnormal conditions during backfilling. Compaction equipment selection shall be controlled, ensuring appropriate type (plate compactor, roller, etc.) is used based on trench width and soil type. Vibration monitoring shall be considered when working near sensitive structures or utilities to prevent damage. Dust monitoring and environmental compliance records shall be maintained in line with CEMP requirements. Access control should be enhanced in active compaction zones to prevent entry of unauthorized personnel. Dedicated banksman shall be assigned for each operating equipment to ensure safe coordination and movement. Stop-work criteria shall be clearly defined, including: -Detection of pipeline movement or damage -Excessive settlement or uneven compaction				

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				L	S	IRR			L	S	RRR	
							Work areas shall be maintained in a clean and organized condition to eliminate trip hazards and ensure safe movement of workers and equipment. Pre-task briefings shall be conducted prior to each shift to communicate hazards, control measures, and emergency procedures. Continuous supervision shall be maintained by Site Engineers, Foremen, and HSE personnel to ensure compliance with safety requirements and immediate correction of unsafe practices. Mandatory PPE including helmet, safety shoes, gloves, high-visibility vest, and dust masks (where required) shall be worn at all times.	-Unstable trench condition -Equipment malfunction -Excessive dust or poor visibility Post-compaction inspection and approval shall be conducted before proceeding to subsequent construction activities. Regular HSE and QA/QC inspections and audits shall be carried out to ensure compliance with method statement and project specifications.				
14.	Use of Power Tools	Untrained users; defective or damaged tools; unguarded rotating parts; electrical shock; loose connections; poor cable management; flying debris; noise exposure; vibration (HAVS); failure of safety devices (dead man switch)	Minor to major injury, electric shock, burns, eye injury, hearing loss, musculoskeletal disorders	3 Rare	4 Major	12 High	Only trained and authorized personnel shall be permitted to operate power tools, and competency shall be verified prior to use. All power tools shall be selected based on suitability for the intended task and used strictly in accordance with manufacturer instructions. All tools shall undergo regular inspection and maintenance, including daily pre-use checks and periodic inspection with color coding system to ensure integrity. Any defective or damaged tools shall be	Toolbox authorization system shall be implemented where only approved and tagged tools are issued to authorized users through tool control register. Portable Appliance Testing (PAT) / electrical inspection records shall be maintained and verified periodically for all electrical tools and extension leads. Double insulation or low-voltage (110V) tools shall	1 Remote	4 Major	4 Major	Supervisor /Storekeeper / Foreman / HSE

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				L	S	IRR			L	S	RRR	
							immediately removed from service, tagged, and replaced. All rotating parts and moving components shall be fitted with proper guards which shall not be removed or bypassed under any circumstances. Safety devices such as dead man switches shall be tested prior to use to ensure functionality. Electrical safety shall be ensured by connecting tools through ELCB-protected distribution boards with proper earthing. All cables and connections shall be inspected to ensure they are free from damage and routed safely to prevent contact with water, sharp edges, or vehicle movement. Cable management shall be maintained to prevent trip hazards, and cables shall be elevated or protected where required. Work areas should be kept clean and organized to ensure safe operation. Exposure to noise and vibration shall be minimized through proper tool selection, maintenance, and limitation of exposure time. Job rotation shall be implemented where necessary to reduce risk of HAVS and fatigue. Flying debris and sparks shall be controlled through proper use of guards and safe positioning of	be used where applicable to reduce risk of electrical shock. Use of Residual Current Device (RCD/ELCB) testing schedule shall be implemented, and trip functionality shall be verified regularly. Dedicated tool inspection checklist shall be implemented and recorded daily before use to ensure all safety features are functional. Anti-vibration tools and ergonomic design tools shall be selected where practicable to reduce exposure to HAVS and musculoskeletal disorders. Work-rest cycle shall be implemented for operators exposed to vibration and repetitive tasks to control fatigue and long-term injury. Tool guarding verification shall be included in inspection to ensure guards are correctly fitted and suitable for the specific task. Dust extraction or suppression system shall be				

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				L	S	IRR			L	S	RRR	
							workers. Work shall be carried out in a controlled area with safe distance maintained from other personnel. Pre-task toolbox talks shall be conducted to communicate hazards, safe operating procedures, and emergency actions. Continuous supervision shall be maintained by Supervisor and HSE personnel. Mandatory PPE including safety goggles/face shield, gloves, hearing protection, helmet, and safety shoes shall be worn at all times. Additional PPE such as dust masks or respirators shall be used where required.	used for cutting/grinding operations to minimize airborne particles and respiratory exposure. Work area isolation (barricading or screens) shall be implemented for high-risk activities such as grinding to protect other workers from flying debris. Secure positioning of workpiece (clamps or vice) shall be ensured before operation to prevent sudden movement and loss of control. Cable routes shall be implemented in congested areas to avoid damage from vehicles, sharp edges, or water exposure. Stop-work criteria shall be clearly defined, including: -Damaged or defective tool identified -Failure of safety device (guard / dead man switch) -Electrical fault or exposed wiring -Excessive vibration or abnormal noise -Unsafe working conditions or poor visibility				

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

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SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
								Regular HSE inspection of tool usage shall be conducted to verify compliance with safety procedures and manufacturer guidelines. Incident/near-miss reporting system shall be implemented for tool-related hazards to improve preventive measures.				
15.	Use of Hand Tools	Sharp edges; struck-by flying chips; improper or unsafe use of tools; defective or damaged tools; repetitive the act; trip hazards; untrained to use chiseling, hammering, cutting activities	Lacerations, eye injury, minor to major injury, musculoskeletal disorders, hand injuries, amputation (in severe cases)	3 Rare	3 Moderate	09 Medium	All hand tools should be inspected prior to use to ensure they are in good condition, free from defects such as cracks, loose handles, or worn-out components. Defective or damaged tools shall be immediately removed from service, clearly identified, and replaced with suitable alternatives. Only appropriate tools shall be selected for each specific task, and misuse of tools (e.g., using spanners as hammers or makeshift extensions) shall be strictly prohibited. Tools shall be ergonomically suitable to minimize strain and improve safe handling. Work should be carried out in a controlled and organized manner, ensuring safe working distance between personnel to avoid struck-by incidents from swinging or flying objects. Sharp tools and cutting edges shall be handled carefully	Tool control and issuance system shall be implemented through store/toolbox register to ensure only approved and serviceable tools are issued to workers. Ergonomic assessment of tools shall be conducted to ensure appropriate size, grip, and design are used to reduce strain and improve handling efficiency. Use of anti-slip grip tools shall be encouraged, especially in wet or oily conditions to prevent loss of control. Workpiece securing methods (clamps, vice, supports) shall be used to stabilize materials during cutting, chiseling, or hammering activities.	1 Remote	3 Moderate	03 Low	Supervisor / Foreman / HSE

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				L	S	IRR			L	S	RRR	
							and directed away from the body at all times. All tools shall be properly stored in designated toolboxes or storage areas when not in use. Tools shall not be left on walkways, platforms, or near excavation edges to prevent trip hazards or falling object risks. Activities such as hammering, chiseling, or cutting shall be performed using appropriate techniques and stable working positions to prevent loss of control or injury. Adequate lighting shall be provided to ensure clear visibility during all operations. Pre-task toolbox talks shall be conducted to brief workers on safe handling practices, hazards, and proper use of hand tools. Workers shall be trained and made aware of correct techniques and limitations of tools. Good housekeeping shall be maintained at all times to ensure safe access and movement within the work area. Continuous supervision shall be ensured by Supervisor and HSE personnel to monitor compliance and correct unsafe practices. Mandatory PPE includes cut-resistant gloves, safety goggles/face shield, helmet, and safety shoes	Dedicated work zones should be established for high-risk activities such as chiseling or hammering to prevent struck-by incidents to nearby workers. Tool carrying methods (tool belts, boxes) shall be used to prevent dropping or unsafe carrying of tools, especially while working at height or near excavation. Storage racks or shadow boards shall be provided to ensure proper organization and easy identification of tools. Periodic refresher training shall be conducted to reinforce correct use, limitations, and safe handling of hand tools. Exposure time to repetitive tasks shall be controlled through job rotation to minimize risk of musculoskeletal disorders. Lighting adequacy shall be verified before starting work, especially in enclosed or low-visibility areas. Stop-work criteria shall be clearly defined, including:				

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				L	S	IRR			L	S	RRR	
							shall be worn at all times to protect against injuries from sharp edges and flying particles.	-Use of defective or damaged tools -Unsafe working posture or condition -Improper tool usage observed -Poor visibility or unstable working surface Regular HSE inspection and audit shall be conducted to verify compliance with safe hand tool practices. Incident and near-miss reporting systems should be implemented for hand tool-related injuries to improve preventive controls.				
16.	Storage of Storm Water Works (GRP & uPVC pipes & Valves) Materials	Pipe rolling; unstable stacking; falling materials; improper storage of chemicals; obstruction of access routes; manual handling during stacking; fire risk (flammable materials); exposure to weather conditions	Crush injury, property damage, musculoskeletal injury, environmental contamination, material damage, fire hazard, project delay	3 Rare	4 Major	12 High	All Storm Water Works materials shall be received, inspected, and stored in designated areas approved by site management to ensure safety and accessibility. Storage areas shall be level, compacted, and capable of supporting the load without settlement or instability. Pipes and large materials shall be stored on timber supports with proper spacing and secured using wedges or stoppers to prevent rolling. Stacking height shall be controlled in accordance with manufacturer recommendations to maintain stability and prevent collapse.	Material storage layout plan shall be prepared and approved, clearly identifying zones for pipes, fittings, chemicals, and hazardous materials. Maximum stacking height limits shall be defined and displayed at storage areas to prevent overloading and collapse. Periodic inspection checklist for storage areas shall be implemented to verify stacking stability, condition of materials, and housekeeping.	1 Remote	4 Major	4 Low	Storekeeper / Site Engineer

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				L	S	IRR			L	S	RRR	
							Different types of materials such as pipes, fittings, chemicals, and construction materials shall be segregated and clearly identified using labels or signage to avoid confusion and unsafe handling. Storage areas shall be located away from traffic routes and excavation edges where practicable, and adequate space shall be maintained for safe movement of personnel and equipment. Access routes shall be kept clear at all times to prevent obstruction and ensure emergency access. Manual handling risks during stacking and retrieval shall be minimized through proper planning, use of mechanical aids where required, and safe lifting techniques. Flammable or hazardous materials shall be stored in designated, well-ventilated areas with appropriate containment measures. Fire prevention measures including availability of fire extinguishers shall be ensured. Materials shall be protected from adverse weather conditions such as excessive heat, rain, or direct sunlight where required to maintain quality and integrity. Good housekeeping shall be maintained to ensure storage areas	Pipe stacking shall be arranged in pyramid or cradle configuration where required to enhance stability and prevent rolling. Protective caps shall be provided on pipe ends to prevent ingress of debris and damage to internal surfaces. Material identification system (tagging/barcoding) shall be implemented to ensure traceability and proper usage of approved materials only. Dedicated chemical storage area with MSDS availability shall be ensured, including segregation of incompatible substances. Secondary containment (bundling / drip trays) shall be provided for liquid chemicals and fuels to prevent environmental contamination. Fire risk assessment shall be carried out for storage areas, and appropriate fire protection measures (extinguishers, signage, no-smoking zones) shall be enforced.				

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				L	S	IRR			L	S	RRR	
							remain organized and free from debris or trip hazards. Pre-task briefings shall be conducted to communicate safe storage and handling procedures. Continuous supervision shall be ensured by Storekeeper, Site Engineer, and HSE personnel to monitor compliance and take corrective action where required. Mandatory PPE including gloves, safety shoes, helmet, and high-visibility vest shall be worn by all personnel involved in material handling and storage activities.	Access control measures shall be implemented to restrict entry to authorized personnel only, especially in hazardous material storage areas. Lighting shall be provided in storage areas for safe handling during early morning, night, or low-visibility conditions. Weather protection measures (covers, sheds) shall be used to protect sensitive materials from UV exposure, heat, and rain. Vehicle movement plan within storage area shall be defined to prevent collision with stored materials. Stop-work criteria shall be clearly defined, including: -Unstable stacking observed -Damaged or deteriorated materials -Blocked access or emergency routes -Chemical leakage or spill -Fire hazard or unsafe storage condition Regular HSE and Store inspection shall be conducted to verify				

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				L	S	IRR			L	S	RRR	
								compliance with storage procedures and safety requirements. Waste and scrap materials shall be segregated and removed regularly to avoid congestion and hazards.				
17.	Operation of Plant and Equipment at Site	Reversing collision; blind spots; struck-by/run-over incidents; overturning due to unstable ground; collision with structures, utilities or other plant; plant interaction with live traffic; operator distraction (mobile phone use)	Fatality, crush injury, amputation, major property damage, service disruption, regulatory penalties	4 Likely	5 Catastrophic	20 Very High	All plant and equipment shall be operated only by trained, licensed, and authorized operators with valid certification. Operator competency shall be verified prior to deployment, and site-specific induction shall be provided. Daily pre-start inspections shall be carried out for all plant and equipment to ensure safe working condition, including checks on brakes, steering, lights, alarms, tires, and hydraulic systems. Any defects shall be reported and rectified before use. Reversing and movement of plant shall be controlled through deployment of trained banksman/flagman to guide operators, especially in blind spot areas and congested zones. No reversing shall be carried out without proper guidance. Plant routes and pedestrian walkways shall be clearly defined and segregated using barriers, markings, and signage to prevent	Plant authorization and allocation systems shall be implemented to ensure only approved operators are assigned to specific equipment. Plant inspection and maintenance records (logbook) shall be maintained for each equipment, including daily, weekly, and periodic checks. 360° cameras / proximity warning systems shall be installed on critical plant where it is feasible to eliminate blind spots and enhance operator awareness. Seat belt interlock system shall be enforced for all operators, and operation without seat belts shall be strictly prohibited.	1 Remote	5 Catastrophic	05 Medium	Logistics Supervisor / Plant Dept / Flagman / HSE

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				L	S	IRR			L	S	RRR	
							Interaction between workers and moving equipment. Designated crossing points shall be established and controlled. Speed limits shall be strictly enforced across site (e.g., 20 km/h), and speed control measures shall be implemented where required. Operators shall be briefed on site traffic rules and movement plans. All plant shall be equipped with functional safety features including reverse alarms, flashing beacons, mirrors, and, where applicable, proximity warning systems. Ground conditions shall be assessed prior to plant operation to ensure stability, especially near excavation edges or soft ground. Safe distance shall be maintained from trenches to prevent collapse or overturning. Use of mobile phones or any form of distraction during operation of plant shall be strictly prohibited and enforced through supervision and disciplinary control. Work near live traffic shall be coordinated with Traffic Management Team and controlled in accordance with approved traffic arrangements. Pre-task briefings shall be conducted to communicate hazards,	Reversing cameras and audible alarms shall be verified functional prior to operation and included in daily inspection checklist. Dedicated plant movement plans (site logistics plan) shall be implemented to control routes, parking, loading/unloading areas, and interaction with other activities. Physical segregation using rigid barriers shall be enhanced in high-risk areas (not only signage or tape). Pedestrian exclusion zones shall be clearly demarcated, and no pedestrian movement shall be allowed in active plant operation zones. Plant parking and isolation procedures shall be implemented, including engine off, key removal, and wheel chocking when not in use. Permit-to-operate (PTO) shall be implemented for critical plant such as cranes, excavators near utilities, or work near live traffic.				

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				L	S	IRR			L	S	RRR	
							movement plans, and emergency procedures. Continuous communication shall be maintained between operators, banksmen, and supervisors. Mandatory PPE including high-visibility vest, helmet, and safety shoes shall be worn by all personnel. Continuous supervision shall be ensured by Logistics Supervisor, Site Engineer, and HSE personnel to monitor compliance and take immediate corrective action where required	Night operation control measures shall include adequate lighting, reflective markers, and enhanced supervision. Fatigue management system shall be implemented for operators, including work-rest cycle and monitoring of working hours. Alcohol and drug policy enforcement shall be strictly implemented, including -random checks where required. -Communication system (radio communication) shall be used between operator, banksmen, and supervisor for real-time coordination. -Stop-work criteria shall be clearly defined, including: -Poor visibility or unsafe weather conditions -Failure of safety devices (alarm, brake, lights) -Unsafe ground condition or edge instability -Unauthorized personnel entering plant zone -Operator fatigue or unsafe behavior Regular HSE inspection shall be conducted to verify compliance with safety				

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				L	S	IRR			L	S	RRR	
								requirements and manufacturer guidelines. Near-miss and incident reporting system shall be implemented for plant-related hazards to improve preventive measures.				
18.	Working Adjacent to Live Traffic	Hit by moving vehicles; unsafe or improper diversion setup; uncontrolled interface between traffic and work zone; premature or unauthorized road closure; inadequate or damaged traffic control devices; poor visibility; driver error or speeding	Fatality, serious injury, public accident, property damage, regulatory penalties, work suspension, reputational impact	4 Likely	5 catastrophic	20 Very High	No work shall commence without approved Traffic Management Plan (TMP), valid RTA ROW permit, and all required authority approvals. Traffic diversions shall be implemented strictly in accordance with RTA Work Zone Traffic Management Manual and approved drawings. Work zones shall be properly designed and established with adequate buffer distances based on road speed and site conditions. Physical segregation between live traffic and work area shall be ensured using approved barriers such as interlocked concrete or water-filled barriers. All traffic control devices including warning signs, directional signs, cones, delineators, flashing beacons, and road markings shall be installed, maintained, and inspected regularly to ensure visibility and effectiveness. Advance warning signage shall be placed at required distances as per RTA standards.	Stage-wise traffic diversion implementation plan shall be verified on site prior to execution to ensure alignment with approved TMP drawings. Pre-opening joint inspection (TTM + HSE + Engineer) shall be conducted before releasing any traffic diversion to confirm all control measures are in place. Deployment of additional TMA units shall be considered for high-speed roads, multiple lane closures, or complex diversion stages. Crash-rated barriers or enhanced protection systems shall be used in high-risk zones to provide improved protection against errant vehicles. Real-time traffic monitoring (CCTV or spotters) shall be	1 Remote	5 catastrophic	05 Medium	TTM Manager / Site Eng / Supervisor / HSE

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				L	S	IRR			L	S	RRR	
							Third-party certified Truck Mounted Attenuator (TMA) shall be deployed during installation and removal of traffic diversions and at critical locations to protect workers from errant vehicles. Competent Traffic Management Team including Traffic Engineer, Supervisor, and trained operatives shall be deployed to monitor and manage traffic flow. Flagmen/banksmen shall be positioned at critical points to control vehicle movement and ensure safe interaction with site activities. Speed control measures including regulatory signage and traffic calming arrangements shall be implemented within and around the work zone. Drivers and operators shall be briefed on site traffic rules and approved access routes. Night works or low-visibility conditions shall be supported with adequate lighting, reflective signage, and enhanced visibility measures. Daily inspection of traffic arrangements including barrier alignment and signage condition shall be carried out and recorded. Strict control shall be maintained to prevent unauthorized entry of public or workers into live carriageways. Safe pedestrian routes and crossing	implemented where required to observe traffic behavior and respond quickly to unsafe situations. Dedicated traffic marshal at critical points shall be assigned to manage sudden traffic flow changes or unexpected driver behavior. Enhanced illumination system (tower lights, LED arrow boards) shall be used during night works to improve visibility of workers and traffic devices. Reflective road studs / temporary lane markings shall be installed to guide drivers clearly through diversion paths. Periodic inspection checklist (daily / shift-based) shall be implemented to verify condition of signage, barriers, cones, and delineators. Weather monitoring systems shall be implemented, and work shall be suspended during adverse conditions such as fog, sandstorms, or heavy rain affecting visibility.				

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							arrangements shall be clearly defined and controlled. Emergency escape routes for workers shall be identified and kept clear at all times. Communication systems (two-way radios) shall be used between traffic team and site personnel for effective coordination. Pre-task toolbox talks and safety campaigns shall be conducted regularly to reinforce awareness of live traffic hazards. Incident reporting and response procedures shall be established in line with RTA requirements. Mandatory PPE including high-visibility clothing, helmet, safety shoes, and gloves shall be worn by all personnel. Continuous supervision shall be ensured by Traffic Management Team, Site Engineer, and HSE personnel to enforce compliance and implement immediate corrective actions.	Public information and awareness measures (advance notice boards, variable message signs) shall be used where required to inform road users about upcoming diversions. Pedestrian safety enhancement including dedicated walkways, guardrails, and controlled crossing points shall be ensured in public interface areas. Strict access control systems shall be enforced to prevent unauthorized vehicles or personnel entering work zones. Emergency response coordination plan shall be established with Traffic Police and RTA for quick intervention in case of incidents. Stop-work criteria shall be clearly defined, including: -Poor visibility conditions -Missing/damaged traffic control devices -Unsafe driver behavior or excessive speeding -Barrier displacement or failure				

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				L	S	IRR			L	S	RRR	
								-Traffic congestion leading to unsafe conditions Regular joint inspection shall be conducted by Traffic Management Team, HSE, and Consultant to verify compliance with TMP and RTA standards. Incident and near-miss reporting system should be implemented specifically for traffic-related events to improve control measures.				
19.	Lifting Operation (Using Mobile Crane / HIAB)	Load drop; crane overturning; rigging failure; uncontrolled load swing; struck-by suspended load; collision with structures or plant; contact with overhead services; unstable ground; poor communication during lifting	Fatality, crush injury, amputation, major property damage, service disruption, regulatory penalties	4 Likely	5 Catastrophic	20 Very High	Lifting operations shall not be carried out without an approved lifting plan, valid Permit-to-Work, and proper risk assessment. All lifting activities shall be planned considering load weight, lifting radius, ground condition, and surrounding environment. Only certified mobile cranes/HIAB and lifting accessories with valid third-party certification shall be used. All equipment including slings, shackles, hooks, and lifting gear shall be inspected prior to use and properly color coded. Crane setup area shall be assessed to ensure ground stability, and outriggers shall be fully extended on firm and level ground using suitable support (steel plates or timber mats) to prevent overturning.	Detailed lifting plan shall include load chart verification, lifting radius, boom configuration, and ground bearing pressure (GBP) calculations, reviewed and approved prior to execution. Critical lift classification system shall be implemented (e.g., heavy lift, tandem lift, near utilities), requiring additional approval and supervision. Ground bearing capacity assessment shall be conducted and documented to ensure crane stability, especially near excavations or soft ground.	1 Remote	5 Catastrophic	05 Median	Lifting Supervisor / Crane Operator / Rigger / Flagman / HSE

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				L	S	IRR			L	S	RRR	
							Load weight shall be verified and shall not exceed crane capacity. Lifting shall be carried out within safe working limits considering radius and boom configuration. A full exclusion zone shall be established around lifting area, and no personnel shall be allowed under suspended loads. Physical barriers and warning signage shall be provided to restrict access. Competent lifting team including certified operator, rigger, banksman, and lifting supervisor shall be deployed for all lifting activities. Clear communication shall be maintained using standard hand signals or radio communication. Taglines shall be used to control load movement and prevent uncontrolled swinging or rotation. Lifting shall be carried out smoothly without sudden jerks. Overhead obstructions such as power lines, structures, and adjacent works shall be identified and safe clearance maintained at all times. Wind speed shall be monitored continuously, and lifting operations shall be suspended when wind	Pre-lift checklist shall be completed and signed before each lifting operation, covering crane condition, rigging, communication, weather, and exclusion zone. Independent lifting supervisor (third-party or competent authority) shall verify lifting setup for high-risk or critical lifts. Rigging plan shall be prepared, including sling angle calculation, load distribution, and proper selection of lifting accessories. Use of certified lifting points / pad eyes only shall be ensured, and improvised lifting methods shall be strictly prohibited. Anemometer (wind speed measuring device) shall be used for monitoring wind conditions during lifting operations. Load path shall be clearly defined and kept free from obstruction, and lifting shall not be carried out over personnel or occupied areas.				

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							exceeds safe operational limits as per manufacturer guidelines. Pre-lift inspection and briefing shall be conducted before each lifting activity to confirm readiness, roles, and safety requirements. Trial lift shall be carried out for critical or heavy lifts. Mandatory PPE including helmet, gloves, safety shoes, and high-visibility vests shall be worn. Additional fall protection should be used where required. Continuous supervision shall be ensured by Lifting Supervisor, Site Engineer, and HSE personnel to enforce compliance and immediately stop unsafe operations.	Secondary safety measures (e.g., safety slings or backup rigging) should be considered for critical lifts. Barricading shall be enhanced using rigid barriers in high-risk or public interface areas. Dedicated communication systems (radio communication) shall be used for complex lifting operations. Simultaneous operations shall be restricted within lifting zone (no excavation, traffic movement, or other conflicting activities). Permit-to-Lift (PTL) system shall be implemented as part of PTW for critical lifting operations. Stop-work criteria shall be clearly defined, including: -Excessive wind speed -Crane instability or ground settlement -Improper rigging or load imbalance -Communication failure -Entry of unauthorized personnel into exclusion zone				

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				L	S	IRR			L	S	RRR	
								Post-lift inspection and verification shall be carried out to ensure no damage to load, lifting accessories, or surrounding structures. Regular lifting inspection shall be conducted by HSE and lifting supervisor to ensure compliance with lifting standards and procedures.				
20.	Manual Handling	Lifting heavy loads; repetitive movement; awkward posture; sudden strain; carrying materials over uneven ground; improper handling techniques; fatigue	Musculoskeletal disorders, back injury, strains and sprains, fatigue-related incidents, reduced productivity	3 Rare	4 Major	12 High	Manual handling activities shall be minimized through proper planning and use of mechanical aids such as trolleys, wheelbarrows, forklifts, or lifting equipment wherever practicable. Work should be organized to reduce the need for excessive manual lifting and carrying. Loads shall be assessed prior to lifting to determine weight, size, and stability. Team lifting shall be implemented for heavy or bulky materials to distribute load safely among workers. Workers shall be trained in proper manual handling techniques including correct lifting posture (straight back, bent knees), safe carrying methods, and avoidance of twisting movements.	Manual Handling Risk Assessment (task-specific) shall be conducted for high-risk lifting activities to evaluate load weight, frequency, posture, and environmental conditions. Maximum safe lifting limits shall be defined and communicated to workers and loads exceeding limits shall be handled using mechanical aids or team lifting. Use of mechanical lifting aids (trolleys, dollies, lifting frames) shall be enforced wherever it is practical to eliminate or reduce manual effort.	1 Remote	4 Major	4 Low	Supervisor / HSE Officer / Trade Foreman

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				L	S	IRR			L	S	RRR	
							Work areas and access routes shall be maintained in good condition, free from obstacles, uneven surfaces, or slip hazards to ensure safe movement during handling of materials. Handling distances shall be minimized through proper material storage planning and positioning of materials close to point of use. Repetitive tasks shall be controlled through job rotation and adequate rest breaks to reduce fatigue and strain. Pre-task toolbox talks shall be conducted to communicate hazards, safe techniques, and control measures related to manual handling. Supervision should be maintained to monitor safe practices and correct improper handling techniques immediately. Mandatory PPE including gloves, safety shoes, and helmet shall be worn to protect against hand injuries and foot impact during handling activities.	Ergonomic assessment of work tasks and layout shall be carried out to reduce bending, twisting, and overreaching during handling activities. Pre-lifting plan or methods shall be established for heavy or awkward loads, including route planning and placement location. Use of proper gripping techniques and anti-slip gloves shall be ensured to improve control during handling. Workplace layout optimization shall be implemented to reduce carrying distances and avoid unnecessary handling. Handling aids such as rollers or skids shall be used for moving heavy materials over short distances. Fatigue monitoring systems shall be implemented, and supervisors shall ensure				

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SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
								workers are not exposed to excessive repetitive strain. Health surveillance or periodic assessment shall be considered for workers exposed to repetitive manual handling tasks. Lighting adequacy shall be ensured in handling areas to prevent misjudgment of load position or movement. Stop-work criteria shall be clearly defined, including: -Load exceeds safe lifting capacity -Unstable or awkward load condition -Unsafe posture or worker fatigue observed -Slippery or obstructed access route -Inadequate manpower for team lifting Near-miss and incident reporting systems shall be implemented for manual handling-related injuries to improve preventive measures.				

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				L	S	IRR			L	S	RRR	
								Regular HSE inspection shall be conducted to verify compliance with safe manual handling practices.				
21.	Interface with Public & Other Contractors	Unauthorized public entry into work zone; pedestrian-vehicle conflict; overlapping contractor activities; conflicting access/traffic arrangements; falling objects; poor communication; unclear work boundaries; unsafe handover areas	Fatality, injury to public/workers, property damage, public complaints, regulatory penalties, work delays, reputational damage	3 Rare	4 Major	12 HIGH	Work areas shall be clearly defined, secured, and segregated from public and adjacent contractor zones using physical barriers such as fencing, hoarding, concrete barriers, and warning tapes. No unauthorized access shall be permitted into active work zones. All access points shall be controlled and monitored, with clear signage indicating restricted areas, safe routes, and hazard warnings. Pedestrian pathways shall be safely diverted with proper guidance, lighting, and protective measures in accordance with approved Traffic Management Plan (TMP) and RTA requirements. Formal interface coordination shall be established between all contractors, including clear definition of work boundaries, responsibilities, access routes, and sequencing of activities. Interface risks shall be identified in advance and controlled through coordinated planning and approvals.	Permit-to-Work (PTW) coordination system shall be implemented to control simultaneous high-risk activities across contractors. Joint risk assessments (JRA) shall be conducted for overlapping activities to identify combined risks and implement shared control measures. Public protection measures shall be enhanced through installation of rigid hoarding, covered walkways, and protective barriers in high-exposure areas. Real-time monitoring (CCTV / spotters) shall be implemented in high-risk interface zones to detect unsafe conditions and unauthorized access. Public awareness measures such as information boards, advance warning signage, and contact details for complaints shall be displayed at site boundaries.	1 Remote	4 Major		Community Liaison / HSE / Construction Manager

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				L	S	IRR			L	S	RRR	
							Daily coordination meetings and look-ahead planning sessions shall be conducted to identify potential conflicts and ensure safe execution of simultaneous activities. All interface-related hazards shall be communicated through toolbox talks and site briefings. Traffic management arrangements shall be coordinated across all contractors to ensure consistency and avoid conflicting diversions or unsafe routing. Shared access routes should be clearly defined and controlled. Work near public areas shall be planned to minimize exposure, and additional protective measures such as overhead protection, debris netting, or screens shall be provided where required. Handover areas between contractors shall be formally inspected and verified to ensure they are safe, stable, and free from hazards before transfer of control. No work shall proceed in partially completed or unsafe interface zones. Effective communication systems including radio or designated coordination channels shall be established to ensure real-time	Dedicated Interface Coordinator shall be assigned to manage coordination between contractors and ensure compliance with interface control measures. Handover checklist and permit system shall be implemented for transfer of work zones between contractors, ensuring formal acceptance of safe conditions. Emergency coordination plan shall be established between contractors, including communication with emergency services and clear roles during incidents. Access and egress control plan shall be implemented to manage shared routes and prevent conflict between contractor operations and public movement. Enhanced pedestrian safety measures including guardrails, anti-slip walkways, and controlled crossings shall be provided in public interface areas.				

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							coordination between contractors, supervisors, and traffic teams. Dust, noise, and vibration shall be controlled to minimize impact on surrounding areas, and any public complaints shall be addressed promptly through designated channels. Security personnel shall be deployed where required to control access and prevent unauthorized entry, particularly during non-working hours. Joint inspections and audits shall be conducted regularly by Site Engineers and HSE teams to verify compliance and identify improvement actions. Pre-task briefings shall be conducted to communicate interface risks, control measures, and emergency procedures. Continuous supervision shall be ensured by Site Engineer, Interface Coordinator, and HSE personnel to enforce compliance and take immediate corrective action where required. Mandatory PPE including high-visibility vest, helmet, and safety shoes shall be worn at all times. Additional protective measures shall be implemented based on	Environmental monitoring (dust, noise, vibration) should be recorded and controlled in compliance with CEMP to minimize public disturbance. Stop-work criteria shall be clearly defined, including: -Unauthorized public entry into work zone -Conflicting contractor activities without coordination -Failure of barricading or access control -Unsafe handover conditions between contractors -Public complaint indicating unsafe condition Regular joint inspections and interface audits shall be conducted with all contractors and consultant representatives. Incident and complaint management system shall be implemented, with proper recording, investigation, and corrective actions.				

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							site conditions and proximity to public areas.					
22.	Refueling of Plant & Machineries	Fire or explosion due to ignition sources; fuel spillage and leakage; exposure to fuel vapors; static electricity discharge; environmental contamination; improper storage of fuel; refueling near hot surfaces or running engines	Fatality, burns, fire damage, environmental pollution, regulatory penalties, property damage	4 Likely	5 Catastrophic	20 Very High	Refueling activities shall be carried out only in designated and approved fueling areas away from ignition sources, open flames, and high-traffic zones. No refueling shall be conducted without proper supervision and measures in place. All plants and equipment shall be switched off and allowed to cool down prior to refueling to prevent ignition of fuel vapors. Refueling shall not be carried out near hot surfaces, electrical sources, or ongoing hot work activities. Fuel shall be stored in approved containers or tanks placed within bunded areas with adequate capacity to contain potential spills. Containers shall be clearly labeled and maintained in good condition to prevent leakage. Spill kits shall be readily available at fueling locations, and personnel involved in refueling shall be trained in spill response procedures. Any spillage shall be immediately contained, cleaned, and disposed of in accordance with environmental requirements. Proper earthing/grounding arrangements shall be ensured during fuel transfer to prevent static electricity buildup and discharge.	Dedicated Refueling Procedure / SOP shall be developed and implemented, including step-by-step controls, emergency actions, and responsibilities. Permit-to-Refuel system shall be implemented for refueling in sensitive areas or near public/traffic interface. Fuel storage and refueling area layout plan shall be prepared, clearly indicating safe distances from ignition sources and access control. Use of intrinsically safe / anti-static equipment shall be ensured to minimize ignition risk during fuel transfer. Static bonding and grounding cables shall be connected before fuel transfer and verified prior to operation. Fuel level monitoring devices / auto shut-off nozzles shall be used	1 Remote	5 Catastrophic	5 Medium	Site Supervisor / Logistics / HSE / Plant Dept

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							Refueling hoses and equipment shall be inspected regularly to ensure integrity. Smoking and use of mobile phones shall be strictly prohibited within fueling areas, and warning signage shall be prominently displayed. Refueling operations shall be carried out by trained personnel using proper techniques to avoid overfilling, splashing, or vapor release. Drip trays shall be used where required to prevent ground contamination. Fire extinguishers (foam, DCP) shall be available at fueling locations and inspected regularly to ensure readiness. Emergency response procedures shall be in place to address fire or spill incidents. Pre-task briefings shall be conducted to communicate hazards and control measures. Continuous supervision shall be ensured by Supervisor and HSE personnel to monitor compliance and enforce safe practices. Mandatory PPE including gloves, safety shoes, helmet, and eye protection shall be worn. Additional protective equipment should be used as required to prevent exposure to fuel.	where practicable to prevent overfilling. Secondary containment (double bunding / drip trays) shall be enhanced to prevent soil and groundwater contamination. Environmental monitoring and recordkeeping shall be maintained for fuel usage, spills, and disposal in compliance with CEMP. Regular inspection checklist for fuel storage and refueling equipment shall be implemented (hoses, valves, containers, bund condition). Designated fire safety zone shall be established around fueling areas with clear demarcation and restricted access. Emergency isolation measures shall be in place, including quick shut-off valves and emergency stop systems. Night refueling control measures shall include adequate lighting and enhanced supervision.				

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								Fuel transport and handling procedures should be implemented to ensure safe delivery and transfer to site storage. Waste fuel and contaminated materials shall be collected, stored, and disposed of through approved environmental procedures. Stop-work criteria shall be clearly defined, including: -Presence of ignition source in fueling area -Fuel leakage or damaged equipment -Absence of spill kit or fire extinguisher -Improper grounding/bonding -Unauthorized personnel in fueling zone Regular HSE and environmental inspection shall be conducted to verify compliance with fuel handling and spill prevention requirements. Incident and spill reporting system shall be implemented with immediate investigation and corrective actions.				

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23.	Environmental Controls (Dust, Noise, Spillage & Vibration) - All Activities	Dust generation from excavation, backfilling, and compaction; excessive noise from equipment; fuel/oil/chemical spillage; vibration from compactors and plant; improper waste handling; uncontrolled runoff	Respiratory irritation, hearing damage, environmental pollution, soil/water contamination, public complaints, regulatory fines, project delay	5 Common	3 Moderate	15 High	All activities shall be carried out in compliance with the approved Construction Environmental Management Plan (CEMP) and Construction Waste Management Plan (CWMP), along with applicable Dubai Municipality and authority regulations. Environmental aspects shall be identified, monitored, and controlled throughout project execution. Dust generation shall be controlled through continuous water spraying at source during excavation, backfilling, and compaction activities. Stockpiles and loose materials shall be covered where required, and vehicle movement shall be controlled to minimize dust emission. Noise levels should be controlled by using well-maintained equipment, proper silencers, and, where required, installation of temporary noise barriers. Work shall be scheduled to minimize disturbance to surrounding areas, especially during sensitive hours. All fuel, oil, and chemical storage shall be maintained in designated bunded areas with proper containment systems. Spill kits shall be readily available, and personnel shall be trained in spill response procedures. Immediate action shall be taken to contain and clean any	Environmental Aspect & Impact Register (EAI) shall be maintained and updated regularly for all site activities. Dedicated Environmental Officer / Coordinator shall be assigned to monitor compliance with CEMP and regulatory requirements. Real-time environmental monitoring devices shall be used where required for dust (PM10/PM2.5), noise levels, and vibration. Vehicle wheel-wash system should be installed at site exit points to prevent mud and dust transfer to public roads. Dust suppression frequency schedule shall be defined and monitored to ensure effectiveness during dry and windy conditions. Use of eco-friendly materials and methods shall be encouraged to minimize environmental impact. Spill prevention plan shall include secondary containment; routine	1 Remote	3 Moderate	3 Low	HSE Officer / Site Engineer / QAQC Engineer

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							spillage to prevent environmental contamination. Vibration exposure shall be controlled through proper equipment selection, maintenance, and limitation of exposure duration. Monitoring shall be conducted where work is carried out near sensitive structures or utilities. Waste materials shall be segregated at source into recyclable, non-recyclable, and hazardous categories and disposed of through approved methods and authorized vendors. Waste accumulation at site shall be avoided through regular removal and proper housekeeping. Runoff and sediment control measures such as silt fences, sandbags, and drainage arrangements shall be implemented to prevent contamination of surrounding areas and water bodies, especially during rain. Regular environmental monitoring for dust, noise, and vibration shall be conducted and recorded in accordance with project requirements. Inspection and audit records shall be maintained for compliance verification. Pre-task briefings should include environmental awareness and	inspection, and immediate reporting system. Waste tracking system (waste manifest) shall be implemented to ensure proper disposal through approved vendors. Segregated waste bins with labeling shall be provided at site to encourage proper waste management. Noise and vibration monitoring near sensitive receptors (residential areas, utilities) shall be conducted and documented. Temporary enclosures / acoustic barriers shall be provided for high-noise equipment where required. Stormwater management plans shall be implemented to control runoff and prevent contamination during rainfall. Environmental incident reporting systems should be implemented with corrective and preventive actions. Stop-work criteria shall be clearly defined, including:				

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							control measures. Public interface shall be managed to address complaints related to dust, noise, or vibration promptly. Mandatory PPE including dust masks, hearing protection, helmet, safety shoes, and gloves shall be used as required. Continuous supervision shall be ensured by HSE personnel to monitor compliance and implement corrective actions where necessary.	-Excessive dust or visibility reduction -Noise exceeding permissible limits -Uncontrolled spill or leakage -Environmental complaints from public -Failure of control measures Regular environmental inspection shall be conducted and records maintained for compliance verification.				
24.	Working in Adverse Weather	Heat stress / heat stroke; dehydration; sandstorm / dust storm (poor visibility); lightning strike; heavy rain / flooding; slippery surfaces; high wind affecting lifting and structures	Fatality, heat exhaustion, respiratory illness, electrocution, slips/falls, drowning, struck-by incidents, property damage	4 Likely	4 Major	16 Very High	Work activities shall be planned and executed considering prevailing and forecasted weather conditions. Daily monitoring of weather forecasts shall be conducted using official sources, and work schedules shall be adjusted accordingly to minimize exposure to adverse conditions. During high temperature conditions, compliance with Dubai Municipality Summer Working Hours regulation shall be strictly enforced, including midday break requirements. Adequate shaded rest areas, drinking water stations, electrolyte supply, and cooling facilities shall be provided at site. Workers shall be monitored for	Weather monitoring system (real-time apps / alerts / anemometer) shall be implemented to track temperature, wind speed, humidity, and visibility continuously. Heat stress index monitoring (WBGT - Wet Bulb Globe Temperature) shall be conducted where required, and work shall be adjusted based on threshold limits. Heat stress management program should be implemented including acclimatization of workers,	1 Remote	4 Major	4 Low	HSE Officer / Site Supervisor / Male Nurse / Project Management

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				L	S	IRR			L	S	RRR	
							signs of heat stress through supervision and buddy system. In case of sandstorms or dust storms, work shall be suspended where visibility is reduced to unsafe levels. Personnel shall be moved to safe shelter, and respiratory protection shall be used where required. All outdoor work shall be stopped immediately during lightning conditions, and personnel shall be evacuated to designated safe areas. No work shall be carried out near metallic structures, cranes, or open areas during lightning risk. During heavy rain, excavation areas shall be protected from flooding through proper drainage arrangements and dewatering systems. Work areas shall be inspected after rainfall to ensure stability of ground, scaffolding, and structures before resuming activities. High wind conditions shall be continuously monitored, especially during lifting operations and work at height. Lifting activities shall	scheduled rest breaks, and hydration monitoring. Buddy systems shall be strictly enforced during high-temperature conditions to identify early signs of heat stress or fatigue. Cooling stations / rest shelters with fans or air cooling shall be provided at strategic locations. Emergency medical response for heat illnesses shall be established, including trained first aiders and immediate cooling procedures. Dust monitoring and suppression systems shall be enhanced during sandstorm or dry conditions to reduce respiratory exposure. Face fit-tested respiratory protection shall be used where dust levels are high.				

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				L	S	IRR			L	S	RRR	
							be suspended when wind speed exceeds safe operational limits. Loose materials and equipment shall be secured to prevent displacement. Slippery surfaces caused by rain or water accumulation shall be controlled through housekeeping, use of anti-slip measures, and restriction of access where required. Pre-task briefings shall be conducted daily to communicate weather-related risks and control measures. Emergency response procedures should be established for heat illness, lightning, flooding, and wind-related incidents. Mandatory PPE including helmet, high-visibility vest, and safety shoes should be worn. Additional PPE such as cooling vests, dust masks, goggles, and rain protection shall be used based on weather conditions. Continuous supervision shall be ensured by Site Engineer and HSE personnel to monitor weather conditions, enforce stop-work criteria, and implement immediate	Temporary securing of loose materials and structures shall be carried out before onset of strong winds. Drainage and stormwater management plans shall be implemented to prevent water accumulation and site flooding. Temporary covers and protection for electrical systems and materials shall be provided during rain. Inspection checklist after adverse weather event shall be implemented before resuming work (covering excavation stability, scaffolding, plant, and electrical systems). Night and low-visibility operations shall be restricted or enhanced with additional lighting and reflective controls.				

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							corrective actions where necessary.	Work rescheduling strategy shall be implemented to shift high-risk activities to safer time periods (early morning/evening). Stop-work criteria shall be clearly defined, including: -Temperature exceeding safe working limits -Wind speed beyond safe lifting/working limits -Visibility reduced below safe level -Lightning detected in vicinity -Flooding or unstable ground conditions Daily weather briefing shall be included in toolbox talks to inform workforce about expected conditions and precautions. Regular HSE inspection and monitoring records shall be maintained for weather-related controls and compliance.				

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25.	Testing & Commissioning for Storm Water Works Pipeline	Pipe rupture during pressure testing; sudden release of pressurized water; failure of joints or fittings; uncontrolled movement of hoses; water leakage/flooding; equipment malfunction; electrical hazards (pumps/control panels)	Fatality, serious injury, impact injury, flooding of excavation, property damage, environmental impact, system failure	4 Likely	5 Catastrophic	20 Very High	Testing and commissioning activities shall be carried out only after completion of installation work and approval of inspection by relevant authority/consultant. No testing shall commence without approved Method Statement, testing procedure, and valid Permit-to-Work. All pipelines, joints, valves, and fittings shall be visually inspected and verified prior to testing to ensure proper installation and readiness. Pressure testing shall be carried out in a controlled and gradual manner as per approved procedure and manufacturer specifications. Pressure gauges and testing equipment shall be calibrated and in good working condition. Test pressure shall not exceed specified limits, and continuous monitoring shall be maintained throughout the testing process. A full exclusion zone shall be established around the testing area, and all non-essential personnel shall be restricted. Personnel shall be positioned at safe locations away from	Detailed testing and commissioning procedure (ITP-based) shall be reviewed and approved, including test pressure, duration, acceptance criteria, and sequence. Hydrostatic pressure test calculation sheet shall be prepared and verified to ensure test pressure is within safe limits of pipeline and fittings. Calibration certificates of pressure gauges and testing equipment shall be verified and maintained on site. Dual pressure gauge system shall be used (primary + secondary gauge) to cross-check readings and avoid false indications. Gradual pressurization in stages (incremental loading) shall be implemented with hold points to monitor system response.	1 Remote	5 Catastrophic	5 Medium	QAQC Engineer / Site Engineer / HSE Officer

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							pressurized lines and potential failure points. All connections, hoses, and fittings shall be properly secured to prevent sudden disconnection or movement under pressure. Emergency arrangements shall be in place to immediately release pressure in case of abnormal conditions. Testing shall be stopped immediately if any leakage, deformation, or unusual behavior is observed. Water discharge during testing shall be controlled and directed to approved drainage points to prevent flooding or environmental impact. Dewatering arrangements shall be available where required. Electrical systems including pumps and control panels shall be properly installed, insulated, and protected from water exposure. Only authorized personnel shall operate electrical equipment. Pre-task briefings shall be conducted to communicate testing procedures, hazards, and emergency response actions. Continuous supervision shall be	Test area barricading shall be enhanced using rigid barriers, especially in high-risk or public interface areas. Pressure relief valve or emergency release systems shall be installed to safely release excess pressure in case of abnormal conditions. All personnel shall be positioned outside the line of fire, and no work shall be allowed directly in front of pressurized joints, caps, or fittings. Dedicated communication system (radio) shall be used between testing team members to ensure coordinated operation. Leak detection and monitoring system shall be implemented during testing to identify weak points early.				

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							ensured by Site Engineer, QA/QC Engineer, and HSE personnel. Mandatory PPE including helmet, safety shoes, gloves, eye protection, and high-visibility vest shall be worn. Additional PPE shall be used based on site conditions and risk exposure.	Water source and discharge capacity shall be verified prior to testing to prevent uncontrolled flooding. Electrical safety verification checklist shall be implemented for pumps and control panels before operation. Environmental protection measures shall be ensured to prevent discharge of contaminated water into surrounding areas. Post-test inspection and documentation shall be completed, including pressure records, leakage observations, and acceptance reports. Permit-to-Test system shall be implemented as part of PTW for controlled execution of testing activities. Stop-work criteria shall be clearly defined, including:				

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								-Sudden pressure drop or abnormal rise -Leakage, deformation, or joint failure -Equipment malfunction -Unsafe discharge or flooding Unauthorized personnel entering test zone Regular inspection by QA/QC and HSE shall be conducted to verify compliance with testing procedure and safety requirements.				
26.	Emergency Arrangement	Fire outbreak; medical emergency; excavation collapse; traffic incident near work zone; electrical shock; flooding due to dewatering failure; delayed evacuation or response	Fatality, major injury, property damage, public impact, regulatory penalties, work suspension	5 Common	4 Major	20 Very High	No work activity shall commence without an approved Emergency Response Plan (ERP) in place, communicated to all personnel, and available at site. The ERP shall address all potential emergency scenarios relevant to Storm Water Works including fire, medical emergencies, excavation collapse, utility incidents, and flooding. Emergency assembly points shall be clearly identified, properly signposted, and kept free from obstruction at all times. Access and egress routes shall be	Site-specific Emergency Organization Chart shall be displayed at prominent locations showing roles (Incident Controller, Fire Warden, First Aider, Rescue Team). Emergency response equipment register shall be maintained and periodically inspected (fire extinguishers, stretchers, rescue kits, spill kits). Dedicated emergency vehicle / standby	1 Remote	4 Major	4 Low	Project Manager / Site Eng / Supervisor / HSE

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				L	S	IRR			L	S	RRR	
							maintained clearly to ensure immediate evacuation of personnel. Emergency contact numbers including Civil Defense, ambulance, police, hospital, and project emergency contacts shall be displayed prominently at site. Coordination shall be established with nearest emergency services to ensure rapid response. Fire extinguishers appropriate to risk (CO₂, DCP, foam) shall be strategically placed at site and inspected regularly to ensure readiness. Spill kits shall be available at designated locations to manage fuel or chemical spills. Trained first aiders shall be available during all working hours and fully equipped first aid kits along with eyewash stations shall be provided at accessible locations. Specific rescue plans shall be developed for high-risk activities such as excavation works, confined spaces, and lifting operations. Rescue equipment	arrangement shall be ensured for quick evacuation of injured personnel. Emergency drill schedule (monthly / quarterly) shall be planned and tracked, with lessons learned documented and implemented documented and learned. Emergency muster roll-call systems shall be implemented to account for all personnel during evacuation. Coordination with nearby hospitals / clinics shall be established, including route mapping and emergency response time verification. Confined space and excavation rescue equipment (tripod, harness, lifeline) shall be pre-positioned for high-risk activities.				

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SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							shall be readily available and maintained. Emergency communication systems including radios and mobile phones shall be available and tested daily to ensure effective coordination during emergencies. Regular emergency drills (fire, evacuation, excavation rescue, medical response) shall be conducted, recorded, and evaluated to verify effectiveness of emergency arrangements and identify areas for improvement. Supervisors and key personnel shall be trained in emergency response procedures, reporting protocols, and escalation processes. Immediate reporting and response shall be ensured in case of any incident. Pre-task briefings shall include emergency awareness and response procedures. Continuous monitoring shall be carried out by Site Engineer and HSE personnel to ensure readiness and compliance. Stop-work authority shall be exercised immediately in	Emergency signage and evacuation route maps shall be displayed in multiple languages for workforce understanding. Alarm system (audible siren / whistle / horn) shall be established and tested regularly for emergency communication. Emergency simulation exercises (scenario-based) shall be conducted for different cases such as fire, collapse, and flooding. Stop-work criteria shall be clearly defined, including: -Fire outbreak or smoke detection -Excavation instability or collapse -Electrical hazard or shock incident -Flooding or system failure -Any uncontrolled emergency situation				

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR DEWA- WD STORM WATERNETWORK										DATE: 23-03-2026		
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							case of any serious unsafe condition or emergency situation.	Incident command system (ICS) shall be implemented to manage emergency response in structured manner. Emergency reporting and investigation systems shall be implemented to analyze response effectiveness and improve preparedness.				
27.	Lighting for Night Work	Inadequate illumination leading to slips / trips / falls; struck-by moving plant / vehicles; poor visibility causing collision; electrical shock / electrocution from faulty lighting; trip hazard from trailing cables; glare / eye strain; fire from overloaded circuits	Fatality; serious injury; crush injury; electrical burns; fire / property damage; regulatory fines	4 Likely	5 Catastrophic	20 Very High	No night work permitted without adequate lighting setup approved by HSE and confirmed in the Method Statement. No use of temporary lighting without Electrical PTW. Lighting levels maintained at a minimum of 200 lux in work areas, 50 lux in access routes (per DM CoCSP). LED floodlights / tower lights used to avoid glare. Cables routed safely, protected from damage and water. Overload protection devices installed on distribution boards. Electrical panels and generators placed in secure, bunded areas away from water and traffic. Live cables clearly marked, secured, and	Additional tower lights / portable floodlights for blind spots or high-risk zones. Daily photo records of lighting setup and lux level measurements. Immediate replacement / repair of any faulty light or cable. PPEs reinforced through daily pre-shift checks.	1 Remote	5 Catastrophic	5 Medium	Electrical Engineer / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

DATE: 23-03-2026

RISK ASSESSMENT FOR DEWA- WD STORM WATERNETWORK

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							covered with conduits or mats to prevent tripping. All temporary lighting installed and tested by competent certified electrician. Pre-use inspection and daily visual checks of lights, cables, and earthing. PTW system enforced. Emergency lighting (battery backup) provided in key areas. Pre-shift toolbox talks on night work hazards, visibility, and cable management. Mandatory high-visibility clothing, safety helmet, safety boots, and eye protection (to reduce glare). All personnel inducted on night work lighting requirements and "no tampering" policy. Strict supervision by Electrical Engineer / HSE. Stop-work authority enforced for inadequate lighting or faults.					

THE END

Appendix B: Inspection & Test Plan

Title:	Method Statement for DM - Storm Water Network			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-UT-MSRA-022	Rev. No.:	00	Page No.:	Page 45 of 47

INSPECTION & TEST PLAN (ITP) for STORM WATER NETWORK

Client	M/s. Roads & Transport Authority (RTA)	Project	R1181-2 - IMPROVEMENT OF AL SAFA ROAD			
Consultant	M/s. Parsons	Discipline	Utilities			
		ITP Ref. No.	CCECC-R1181-2-CC-UT-ITP-022	Rev	00	
Contractor	M/s. China Civil Engineering Construction Corporation – Dubai	Sub Contractor	-	Date	24-Mar-26	

SN	Inspection / Test to be carried out	Frequency of Tests / Inspections	Acceptance Criteria	Applicable Specification / Standard	Control Data / Forms	Inspection Method	Responsibility			
							Contractor / Subcontractor	TPL	Consultant	Client / Authority
1	Pre Approvals / Documentation									
1.1	NOC / Work permit from Relevant Authorities	Prior to start	Approved NOC Documents and comply all NOC conditions	Project Specification Vol-2 / Part 2 / Service works Clause 2/1 to 2/5	NOC Approval	Review	E	N/A	R	A
1.2	Shop Drawing Approvals	Prior to start	Review and Approved by Engineer	Project Specification Vol-2 / Part 2 / Service works Clause 2/1 to 2/5	Submittal approval	Review	E	N/A	R/A	A
1.3	Pre-Qualification Approvals	Prior to start	Review and approved by Engineer	Project Specification Vol-2 / Part 2 / Service works Clause 2/1 to 2/5	Submittal approval	Review	E	N/A	R/A	A
2	Pre-Works									
Pipe Laying										
2.1	Inspection of Materials Receipt at site	As per the Delivery schedule	Code A or B with copliance to conditions	As per Approved MAR/ project specification / manufactures recommendations	NMR	Site Inspection	E	N/A	W/A	S
2.2	Survey & Settingout	Prior to start of works	As per Drawing	As per Project Standard Specification	Survey Data	Survey	E	N/A	W	S
2.3	Excavation and Formation preparation	prior to pipe laying	Shall be compacted to 95% of MDD	As per Project Standard Specification	Test Reports	FDT	E	N/A	W	S
2.4	Installation of the pipes Main Line and Land Drain	Each location	all pipes shall be inspected before installation	As per Project Standard Specification	RFA	Visual Inspection	E	N/A	W/A	S
2.5	Formwork, Placement of concrete and Curing for concrete encasement	Each location	Slump as per approved mix design. Concrete temperature < 30 dg C air temperature <40.	As per Project Standard Specification	RFA	Site Inspection	E	N/A	W/A	S

INSPECTION & TEST PLAN (ITP) for STORM WATER NETWORK

Client		M/s. Roads & Transport Authority (RTA)	Project	R1181-2 - IMPROVEMENT OF AL SAFA ROAD						
Consultant		M/s. Parsons	Discipline	Utilities						
			ITP Ref. No.	CCECC-R1181-2-CC-UT-ITP-022				Rev	00	
Contractor		M/s. China Civil Engineering Construction Corporation – Dubai	Sub Contractor	-				Date	24-Mar-26	
SN	Inspection / Test to be carried out	Frequency of Tests / Inspections	Acceptance Criteria	Applicable Specification / Standard	Control Data / Forms	Inspection Method	Responsibility			
							Contractor / Subcontractor	TPL	Consultant	Client / Authority
2.6	Application of Bitumen emulsion to concrete encasement	Each location	Concrete surface shall be cleaned and Bituminous paint to be applied.	As per Contract Drawing	RFA	Site Inspection	E	N/A	W/A	S
2.7	Air Test	Each Pipe Line (UPVC and GRP upto 600mm)	Pressure should not fall below 75 mm in 5min Interval	As per Project Standard Specification	RFA	Testing	H	E	W/A	S
2.8	Water Test	Each Pipe Line (UPVC and GRP upto 600mm)	Water consumption in 60 minutes shall not exceed 7.5L/m per 30m of length of pipeline under test.	As per Project Standard Specification	RFA	Testing	H	E	W/A	S
2.9	Deflection Test	Each Pipe Line (UPVC solid pipes only)	Mandrel dia 3% of Nominal Dia. for initial and 5% of Nominal Diameter for long term shall pass through the pipe.	As per Project Standard Specification	RFA	Testing	H	E	W/A	S
		Each Pipe Line (GRP dia. larger than 600mm)	Initial - after backfilling: 2% of internal diameter 6 months after installation: 4% of internal diameter	As per project Standard specification	RFA	Testing	H	E	W/A	S
2.10	Backfilling & Compaction	Each layer	Maximum layer thickness 200mm and each layer shall be compacted to a minimum 95% of MDD.	As per Project Standard Specification	Test Reports	FDT	E	N/A	W/A	S
2.11	Installation of Warning tape	Each location	As shown in the Drawings	As per Project Standard Specification	RFA	Visual Inspection	E	N/A	W/A	S

INSPECTION & TEST PLAN (ITP) for STORM WATER NETWORK

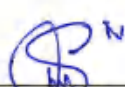
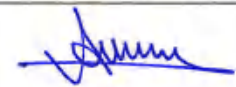
Client	M/s. Roads & Transport Authority (RTA)	Project	R1181-2 - IMPROVEMENT OF AL SAFA ROAD							
Consultant	M/s. Parsons	Discipline	Utilities							
		ITP Ref. No.	CCECC-R1181-2-CC-UT-ITP-022					Rev	00	
Contractor	M/s. China Civil Engineering Construction Corporation – Dubai	Sub Contractor	-					Date	24-Mar-26	
SN	Inspection / Test to be carried out	Frequency of Tests / Inspections	Acceptance Criteria	Applicable Specification / Standard	Control Data / Forms	Inspection Method	Responsibility			
							Contractor / Subcontractor	TPL	Consultant	Client / Authority
3	Construction & Installation of Manholes & Gullies									
3. 1	Inspection of Materials Receipt at site	As per the Delivery schedule	Code A or B with compliance to conditions	As per Approved MAR/ project specification / manufactures recommendations	MIR	Site Inspection	E	N/A	R/A	S
3. 2	Formwork, Placement of concrete and Curing for Manhole, Gully and Cover Slab	Each Manhole, Gully and Cover Slab	Slump as per approved mix design. Concrete temperaturae < 32	As per project Standard specification / Approved shop drawings	RFA	Site Inspection	E	N/A	W/A	S
3. 3	Survey & Settingout	Prior to start of works	As per Drawing	As per Project Standard Specification	Survey Data	Survey	E	N/A	W	S
3. 4	Excavation and Formation preparation	Each Manhole & Gully	Shall be compacted to 95% of MDD	As per Project Standard Specification	Test Reports	FDT	E	N/A	W	S
3. 5	Blinding,waterproofing and screed	Each Manhole	Concrete surface shall be cleaned and Bituminous paint to be applied, Overlapping as per manufacturer's recommendations Screed as per approved drawings	As per project Standard specification / Approved shop drawings	RFA	Site Inspection	E	N/A	W	S
3. 6	Installation of Manhole and Gully	Each Manhole & Gully	As per Drawing	As per Project Standard Specification	RFA	Site Inspection	E	N/A	W/A	S
3. 8	Application of Bitumen emulsion and Waterproofing membrane	Each Manhole (Membrane with protection Board) Each Gully (1 Coat Primer followed 2 Coats RBE) Internal Concrete Surfaces (Epoxy paint 3 coats)	Concrete surface shall be cleaned and Bituminous paint to be applied.	As per project Standard specification / Approved shop drawings	RFA	Site Inspection	E	N/A	W/A	S

INSPECTION & TEST PLAN (ITP) for STORM WATER NETWORK

Client	M/s. Roads & Transport Authority (RTA)	Project	R1181-2 - IMPROVEMENT OF AL SAFA ROAD			
Consultant	M/s. Parsons	Discipline	Utilities			
		ITP Ref. No.	CCECC-R1181-2-CC-UT-ITP-022	Rev	00	
Contractor	M/s. China Civil Engineering Construction Corporation – Dubai	Sub Contractor	-	Date	24-Mar-26	

SN	Inspection / Test to be carried out	Frequency of Tests / Inspections	Acceptance Criteria	Applicable Specification / Standard	Control Data / Forms	Inspection Method	Responsibility			
							Contractor / Subcontractor	TPL	Consultant	Client / Authority
3.9	Pipe connection to Manhole & Gully	Each Manhole & Gully	As per Drawing	As per Project Standard Specification	RFA	Site Inspection	E	N/A	W/A	S
3.10	Backfilling & Compaction	Each layer	Maximum layer thickness 200mm and each layer shall be compacted to a minimum 95% of MDD.	As per Project Standard Specification	Test Reports	FDT	E	N/A	W	S
3.11	Installation of Cover Slab	Each Manhole & Gully	As per Drawing	As per Project Standard Specification	RFA	Site Inspection	E	N/A	W/A	S
3.12	Installation of Cover and Frame	Each Manhole & Gully	As per Drawing	As per Project Standard Specification	RFA	Site Inspection	E	N/A	W/A	S
3.13	Insitu Lamination to Liner & Benching Joints and Pipe Connecting to liners	Each Manhole	As per Drawing	As per Project Standard Specification	RFA	Site Inspection	E	N/A	W/A	S
3.14	Installation of Warning tape	Each location	As shown in the Drawings	As per Project Standard Specification	RFA	Visual Inspection	E	N/A	W/A	S

Note (1): H: Hold Point, W: Witness, R: Review, S: Surveillance, A: Approval, N/A: Not Applicable, E: Execute

	ForSub Contractor	For Contractor				Consultant
Name	N/A	Ahmed Ghanem	Syed Abdul Moid	Ahmed Abdalla	Ahmed Elkadi	
Designation	-	Construction Manager	QA/QC Manager	Technical Manager	Deputy Project Manager	
Signature	-					
Date	-	24-Mar-26	24-Mar-26	24-Mar-26	24-Mar-26	

Appendix C: Checklist

Title:	Method Statement for DM - Storm Water Network			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-UT-MSRA-022	Rev. No.:	00	Page No.:	Page 46 of 47

CHECK LIST

Project Name	R1181 - 2 - Improvement of Al Safa Road	Client	M/s. Roads & Transport Authority (RTA)
Engineer	Parsons	Contractor	M/s. China Civil Engineering Construction Corporation - Dubai
Checklist Title:	Storm Water Works	RFA No.	
Works Location / Structure Reference		Related ITP #	CCECC-R1181-2-CC-UT-ITP-022 Rev : 00
		Related MS #	CCECC-R1181-2-CC-UT-MSRA-022 Rev : 00
Related Drawing(s)		Date	

SN	Description of checking to be carried out	Status - Acceptable			Remarks
		Yes	No	NA	
1	PRE APPROVALS				
1.1	NOC's/ Work Permit available.				
1.2	Approved Method Statement available.				
1.3	Ensure all the survey equipment are calibrated.				
2	PRE WORKS				
2.1	Access area and Working area cleared form materials and debris.				
2.2	Survey to Establish Control Points.				
2.3	Identify the borders of site clearance to set as parameters.				
3	EXECUTION OF WORKS				
3.1	Survey work to be done and checked according to approved shop drawing.				
3.2	Bottom levels to be approved as required in project specification.				
3.3	Check pipe lifted with suitable means and laid inside the trench without any damage.				
3.4	Check pipes to be joined together (Socket / Spigot or Coupler Connection) as per manufacturer's recommendation.				
3.5	Check the pipe and pipe joint is cleaned and lubricated and that the rubber gasket is properly inserted in the grove of the socket end of the pipe.				
3.6	Check all pipe ends are fitted with plugs or caps after pipe laying and be removed before proceeding connection.				
3.7	Check pipe laying is done from downstream to upstream with socket end at upstream end of each pipe.				
3.8	Check pipeline laid in trench to have the permissible tolerance with the required level and alignment / line and grade.				
3.9	Pipes and fittings shall be installed & jointed to the required line & level.				
3.10	Gravel bedding with geotextile filter fabric shall be placed for the pipe located permanently beneath groundwater, and as per approved shop drawing or engineer's request.				
3.11	Concrete surround for the pipes as per approved shop drawing shall be supported with concrete setting blacks to prevent settlement during laying.				
3.12	Compaction of pipe surrounds shall be carefully done at the sides to prevent them from moving or misalignment.				
3.13	Air Test				
3.14	Deflection Test				
3.15	Final Water test				
3.16	Backfilling using suitable material in layers of 250mm thickness.				
3.17	Warning tapes shall be installed as indicated in approved shop drawing.				

Remarks:

	For Contractor		For Consultant (Engineer)
	Sub-Contractor Engineer	CCECC Engineer	
Name			
Designation			
Signature			
Date			

Appendix E: Emergency Contact Details

Title:	Method Statement for DM - Storm Water Network			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-UT-MSRA-022	Rev. No.:	00	Page No.:	Page 47 of 47

Emergency Contact Numbers

Contract No.:	R 1181-2	Project Title:	IMPROVEMENT OF AL SAFA STREET
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PROJECT DIRECTOR

Mohsen D Gazzaz	Mobile	052 282 5807
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DEPUTY PROJECT MANAGER

Ahmad Elkadi	Mobile	050 907 0342
Guo Pingcheng	Mobile	058 103 0678

HSE PERSONAL

Mirza Shahabuddin Baig	Mobile	055 933 1082
Ajmal Rashid	Mobile	056 148 7012
Shahid Ahmad	Mobile	050 560 9961
Basharat Karim	Mobile	054 358 9874
Asif Javed	Mobile	052 787 0311

CONSTRUCTION MANAGER

Ahmad Ghanem	Mobile	055 665 1144
Enji Shi	Mobile	054 738 1666

MALE NURSE

VIJAY SINGH	Mobile	058 837 2856
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SITE ENGINEER

Kasim (Utility & Road Workers)	Mobile	056 143 5466
Syed Abdul Kather (Electrical)	Mobile	055 789 3827
Hashim Ali (Structure)	Mobile	055 288 9793
Kishor Pote	Mobile	056 615 5815

SUPERVISOR

Zakir Ali	Mobile	055 168 6375
Gautam Kumar	Mobile	050 587 9443

EMERGENCY SERVICES

POLICE	☎	999
CIVIL DEFENCE	☎	997
AMBULANCE	☎	998
DUBAI MUNICIPALITY	☎	900
POISON CONTROL	☎	800 424
RTA	☎	800 9090
Iranian Hospital - Dubai	NEAREST HOSPITAL	04 344 0250

SITE ELECTRICIAN

Thomachan Thomas (Electrical Supervisor)	Mobile	055 280 4577
Sanjay	Mobile	052 368 7302